

Supplement-Outlier Detection Methods for One-Dimensional Features Based on the Isolation Forest and 2-Means Algorithms

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Tables

Table 1: Average values of TP/FP/FN/TN for the A1 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	5.059 (2.132)	3.112 (5.316)	0.000 (0.000)	91.830 (4.653)
M.II	4.969 (2.184)	12.813 (4.857)	0.140 (0.876)	82.078 (4.432)
M.III	5.045 (2.109)	10.668 (4.896)	0.025 (0.292)	84.261 (4.215)
M.III+BH	2.250 (1.238)	16.500 (5.967)	0.000 (0.000)	81.250 (6.618)
M.IV	5.133 (2.098)	1.597 (12.181)	0.000 (0.000)	93.270 (11.834)
M.V	2.690 (0.980)	7.096 (1.413)	2.391 (1.920)	87.822 (2.158)
M.VI	4.930 (2.154)	13.588 (6.676)	0.141 (0.752)	81.342 (6.127)
M.VII	5.059 (2.132)	3.112 (5.316)	0.000 (0.000)	91.830 (4.653)
M.VIII	5.070 (2.143)	1.975 (13.789)	0.000 (0.000)	92.955 (13.510)
M.IX (MAD)	5.070 (2.143)	3.342 (2.347)	0.000 (0.000)	91.588 (2.734)
M.X (IQR)	5.070 (2.143)	0.588 (0.853)	0.000 (0.000)	94.342 (2.205)

Table 2: Average values of the classification quality measures for the A1 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.969 (0.053)	0.733 (0.285)	1.000 (0.000)	0.806 (0.251)
M.II	0.870 (0.052)	0.290 (0.119)	0.980 (0.127)	0.435 (0.152)
M.III	0.893 (0.049)	0.337 (0.136)	0.997 (0.030)	0.487 (0.159)
M.III+BH	0.835 (0.060)	0.122 (0.059)	1.000 (0.000)	0.212 (0.090)
M.IV	0.984 (0.122)	0.971 (0.135)	1.000 (0.000)	0.977 (0.125)
M.V	0.905 (0.022)	0.276 (0.098)	0.602 (0.252)	0.361 (0.114)
M.VI	0.863 (0.070)	0.288 (0.127)	0.978 (0.109)	0.431 (0.162)
M.VII	0.969 (0.053)	0.733 (0.285)	1.000 (0.000)	0.806 (0.251)
M.VIII	0.980 (0.138)	0.979 (0.138)	1.000 (0.000)	0.980 (0.135)
M.IX (MAD)	0.967 (0.023)	0.617 (0.209)	1.000 (0.000)	0.742 (0.172)
M.X (IQR)	0.994 (0.009)	0.899 (0.141)	1.000 (0.000)	0.940 (0.089)

Table 3: Average values of TP/FP/FN/TN for the A1 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	49.779 (7.105)	12.842 (8.786)	0.048 (1.521)	937.330 (10.425)
M.II	49.492 (7.186)	103.272 (9.244)	0.330 (1.939)	846.906 (10.424)
M.III	49.820 (6.930)	104.772 (6.287)	0.002 (0.063)	845.406 (7.992)
M.III+BH	27.593 (13.988)	10.672 (7.041)	18.842 (14.754)	942.892 (9.308)
M.IV	49.813 (6.926)	0.014 (0.118)	0.009 (0.094)	950.164 (6.928)
M.V	30.477 (1.685)	69.493 (1.978)	19.345 (6.061)	880.685 (6.108)
M.VI	49.737 (7.043)	107.552 (16.292)	0.085 (1.231)	842.626 (16.907)
M.VII	49.779 (7.105)	13.247 (15.751)	0.048 (1.521)	936.926 (16.716)
M.VIII	49.814 (6.925)	0.021 (0.143)	0.008 (0.089)	950.157 (6.928)
M.IX (MAD)	49.822 (6.929)	30.985 (7.290)	0.000 (0.000)	919.193 (8.772)
M.X (IQR)	49.822 (6.929)	4.300 (2.314)	0.000 (0.000)	945.878 (6.924)

Table 4: Average values of the classification quality measures for the A1 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.987 (0.010)	0.801 (0.084)	0.999 (0.031)	0.887 (0.060)
M.II	0.896 (0.011)	0.324 (0.038)	0.993 (0.039)	0.487 (0.046)
M.III	0.895 (0.006)	0.322 (0.036)	1.000 (0.001)	0.486 (0.041)
M.III+BH	0.970 (0.010)	0.725 (0.070)	0.602 (0.301)	0.611 (0.205)
M.IV	1.000 (0.000)	1.000 (0.002)	1.000 (0.002)	1.000 (0.002)
M.V	0.911 (0.006)	0.305 (0.017)	0.621 (0.071)	0.407 (0.018)
M.VI	0.892 (0.017)	0.317 (0.038)	0.998 (0.025)	0.480 (0.046)
M.VII	0.987 (0.016)	0.800 (0.088)	0.999 (0.031)	0.886 (0.064)
M.VIII	1.000 (0.000)	1.000 (0.003)	1.000 (0.002)	1.000 (0.002)
M.IX (MAD)	0.969 (0.007)	0.618 (0.070)	1.000 (0.000)	0.762 (0.054)
M.X (IQR)	0.996 (0.002)	0.920 (0.042)	1.000 (0.000)	0.958 (0.023)

Table 5: Average values of TP/FP/FN/TN for the A2 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	4.967 (2.213)	9.361 (6.858)	0.000 (0.000)	85.672 (5.949)
M.II	4.967 (2.213)	16.787 (5.774)	0.000 (0.000)	78.246 (4.925)
M.III	4.935 (2.209)	16.290 (6.263)	0.000 (0.000)	78.774 (5.388)
M.III+BH	3.684 (1.916)	15.579 (6.686)	0.000 (0.000)	80.737 (5.989)
M.IV	5.000 (2.181)	1.763 (5.575)	0.085 (0.281)	93.153 (5.333)
M.V	2.468 (0.936)	7.645 (1.516)	2.468 (2.070)	87.419 (2.351)
M.VI	4.935 (2.209)	18.742 (6.455)	0.000 (0.000)	76.323 (5.574)
M.VII	4.967 (2.213)	9.361 (6.858)	0.000 (0.000)	85.672 (5.949)
M.VIII	4.403 (2.343)	24.419 (41.733)	0.532 (1.411)	70.645 (42.034)
M.IX (MAD)	4.903 (2.201)	3.355 (2.430)	0.032 (0.178)	91.710 (2.576)
M.X (IQR)	4.694 (2.053)	0.532 (0.953)	0.242 (0.619)	94.532 (2.407)

Figures

Below, we report the p-values from pairwise comparisons of selected classification metrics: Accuracy, Precision, Recall, and F1, for the considered scenarios A1–A3 and B1–B3. The comparisons

Table 6: Average values of the classification quality measures for the A2 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.906 (0.069)	0.410 (0.223)	1.000 (0.000)	0.546 (0.232)
M.II	0.832 (0.058)	0.242 (0.116)	1.000 (0.000)	0.376 (0.150)
M.III	0.837 (0.063)	0.249 (0.123)	1.000 (0.000)	0.384 (0.157)
M.III+BH	0.844 (0.067)	0.220 (0.168)	1.000 (0.000)	0.336 (0.191)
M.IV	0.982 (0.056)	0.884 (0.228)	0.981 (0.065)	0.905 (0.195)
M.V	0.899 (0.025)	0.246 (0.092)	0.571 (0.261)	0.328 (0.116)
M.VI	0.813 (0.065)	0.223 (0.110)	1.000 (0.000)	0.351 (0.146)
M.VII	0.906 (0.069)	0.410 (0.223)	1.000 (0.000)	0.546 (0.232)
M.VIII	0.750 (0.414)	0.752 (0.416)	0.898 (0.215)	0.693 (0.390)
M.IX (MAD)	0.966 (0.024)	0.613 (0.220)	0.994 (0.031)	0.733 (0.185)
M.X (IQR)	0.992 (0.011)	0.913 (0.141)	0.962 (0.090)	0.927 (0.099)

Table 7: Average values of TP/FP/FN/TN for the A2 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	49.766 (6.926)	54.522 (20.555)	0.049 (0.226)	895.663 (19.856)
M.II	48.749 (7.561)	131.581 (21.691)	1.073 (3.174)	818.597 (21.424)
M.III	49.800 (6.920)	142.093 (20.003)	0.022 (0.204)	808.085 (19.190)
M.III+BH	48.002 (6.786)	48.337 (23.534)	1.813 (4.936)	901.848 (22.338)
M.IV	48.401 (6.940)	2.144 (30.116)	1.421 (1.204)	948.034 (30.859)
M.V	27.946 (2.026)	72.092 (2.365)	21.876 (6.292)	878.086 (6.347)
M.VI	49.682 (7.089)	145.410 (25.908)	0.140 (1.697)	804.768 (25.391)
M.VII	49.766 (6.926)	54.436 (20.602)	0.049 (0.226)	895.749 (19.904)
M.VIII	41.689 (14.711)	155.855 (351.589)	8.133 (13.517)	794.323 (352.051)
M.IX (MAD)	49.688 (6.901)	31.059 (7.580)	0.134 (0.364)	919.119 (9.019)
M.X (IQR)	48.786 (6.748)	4.327 (2.506)	1.036 (1.060)	945.851 (7.011)

Table 8: Average values of the classification quality measures for the A2 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.945 (0.021)	0.493 (0.105)	0.999 (0.004)	0.653 (0.095)
M.II	0.867 (0.023)	0.273 (0.044)	0.978 (0.064)	0.425 (0.056)
M.III	0.858 (0.020)	0.262 (0.046)	1.000 (0.004)	0.413 (0.055)
M.III+BH	0.950 (0.022)	0.524 (0.120)	0.968 (0.084)	0.667 (0.098)
M.IV	0.996 (0.030)	0.976 (0.043)	0.971 (0.025)	0.973 (0.034)
M.V	0.906 (0.006)	0.279 (0.020)	0.569 (0.071)	0.373 (0.024)
M.VI	0.854 (0.027)	0.258 (0.043)	0.997 (0.033)	0.408 (0.055)
M.VII	0.946 (0.021)	0.493 (0.105)	0.999 (0.004)	0.653 (0.096)
M.VIII	0.836 (0.348)	0.841 (0.350)	0.836 (0.266)	0.725 (0.370)
M.IX (MAD)	0.969 (0.008)	0.617 (0.073)	0.997 (0.007)	0.760 (0.055)
M.X (IQR)	0.995 (0.003)	0.919 (0.045)	0.979 (0.021)	0.947 (0.025)

were performed using the Wilcoxon test with Holm correction for multiple testing.

In the figures, the following color coding was used: **White** $\rightarrow$ no significant difference, **Red** $\rightarrow$ highly significant difference. The scale represents $-\log_{10}(p)$: 1.3 corresponds to $p = 0.05$, 3 corresponds to $p = 0.001$ and 10 corresponds to extremely significant differences.

Table 9: Average values of TP/FP/FN/TN for the A3 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	2.183 (1.253)	18.916 (7.666)	3.014 (1.919)	75.886 (7.799)
M.II	2.272 (1.261)	24.022 (6.873)	2.946 (1.870)	70.760 (6.983)
M.III	2.296 (1.248)	23.166 (6.489)	2.855 (1.804)	71.684 (6.509)
M.III+BH	2.119 (1.142)	17.888 (6.092)	2.993 (1.801)	77.000 (6.303)
M.IV	2.912 (1.983)	39.487 (36.239)	2.368 (2.381)	55.232 (35.911)
M.V	1.422 (0.637)	9.165 (1.559)	4.017 (2.028)	85.395 (2.533)
M.VI	2.415 (1.326)	26.900 (7.018)	2.747 (1.780)	67.939 (7.108)
M.VII	2.183 (1.253)	18.916 (7.666)	3.014 (1.919)	75.886 (7.799)
M.VIII	4.917 (2.163)	93.642 (11.498)	0.064 (0.581)	1.377 (11.251)
M.IX (MAD)	1.384 (0.602)	4.352 (2.610)	4.025 (2.076)	90.239 (3.103)
M.X (IQR)	1.121 (0.327)	1.338 (1.347)	4.414 (2.127)	93.127 (2.382)

Table 10: Average values of the classification quality measures for the A3 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.781 (0.074)	0.110 (0.060)	0.458 (0.238)	0.168 (0.080)
M.II	0.730 (0.066)	0.088 (0.045)	0.471 (0.236)	0.143 (0.067)
M.III	0.740 (0.062)	0.093 (0.050)	0.481 (0.229)	0.150 (0.071)
M.III+BH	0.791 (0.060)	0.110 (0.055)	0.451 (0.224)	0.170 (0.078)
M.IV	0.581 (0.348)	0.140 (0.149)	0.599 (0.334)	0.172 (0.117)
M.V	0.868 (0.026)	0.136 (0.062)	0.300 (0.170)	0.179 (0.075)
M.VI	0.704 (0.068)	0.084 (0.044)	0.500 (0.234)	0.139 (0.067)
M.VII	0.781 (0.074)	0.110 (0.060)	0.458 (0.238)	0.168 (0.080)
M.VIII	0.063 (0.110)	0.062 (0.108)	0.989 (0.093)	0.098 (0.052)
M.IX (MAD)	0.916 (0.030)	0.299 (0.191)	0.303 (0.188)	0.260 (0.107)
M.X (IQR)	0.942 (0.024)	0.592 (0.300)	0.254 (0.179)	0.307 (0.124)

Table 11: Average values of TP/FP/FN/TN for the A3 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	14.931 (4.763)	130.488 (33.914)	34.891 (6.560)	819.690 (34.433)
M.II	15.685 (4.997)	182.692 (29.924)	34.137 (6.643)	767.486 (30.423)
M.III	20.026 (4.769)	208.681 (29.688)	29.796 (5.703)	741.497 (30.109)
M.III+BH	15.510 (4.653)	139.711 (34.457)	34.315 (6.397)	810.464 (35.003)
M.IV	27.812 (20.823)	484.873 (401.129)	22.129 (21.245)	465.186 (400.689)
M.V	9.578 (2.917)	90.576 (3.250)	40.244 (6.146)	859.602 (6.322)
M.VI	19.245 (4.870)	211.692 (30.778)	30.577 (5.900)	738.486 (31.187)
M.VII	14.954 (4.785)	131.211 (41.146)	34.868 (6.623)	818.967 (41.342)
M.VIII	46.794 (13.392)	889.578 (230.032)	3.074 (11.944)	60.554 (229.697)
M.IX (MAD)	6.703 (2.720)	38.224 (8.337)	43.119 (6.607)	911.954 (10.301)
M.X (IQR)	2.137 (1.155)	6.167 (2.958)	47.934 (6.813)	943.762 (7.380)

Table 12: Average values of the classification quality measures for the A3 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.835 (0.033)	0.105 (0.030)	0.300 (0.088)	0.153 (0.040)
M.II	0.783 (0.028)	0.079 (0.021)	0.315 (0.091)	0.125 (0.033)
M.III	0.762 (0.029)	0.088 (0.020)	0.402 (0.080)	0.144 (0.030)
M.III+BH	0.826 (0.033)	0.102 (0.027)	0.311 (0.084)	0.151 (0.036)
M.IV	0.493 (0.383)	0.113 (0.144)	0.561 (0.410)	0.111 (0.071)
M.V	0.869 (0.007)	0.096 (0.029)	0.192 (0.053)	0.127 (0.036)
M.VI	0.758 (0.030)	0.084 (0.020)	0.386 (0.083)	0.137 (0.030)
M.VII	0.834 (0.040)	0.105 (0.030)	0.300 (0.089)	0.153 (0.040)
M.VIII	0.107 (0.218)	0.101 (0.210)	0.940 (0.232)	0.093 (0.021)
M.IX (MAD)	0.919 (0.010)	0.150 (0.058)	0.135 (0.052)	0.140 (0.051)
M.X (IQR)	0.946 (0.007)	0.278 (0.146)	0.043 (0.023)	0.073 (0.036)

Table 13: Average values of TP/FP/FN/TN for the B1 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	4.175 (1.813)	5.650 (6.694)	0.842 (1.149)	89.333 (5.886)
M.II	4.190 (1.831)	14.291 (5.552)	0.826 (1.069)	80.693 (4.871)
M.III	4.590 (1.979)	12.197 (5.711)	0.414 (0.639)	82.799 (4.785)
M.III+BH	3.122 (1.618)	13.212 (4.835)	0.317 (0.569)	83.349 (4.420)
M.IV	4.331 (1.798)	1.523 (9.953)	0.849 (0.939)	93.297 (9.751)
M.V	3.174 (1.222)	6.846 (1.642)	1.837 (1.590)	88.143 (1.897)
M.VI	4.484 (1.952)	15.202 (7.214)	0.520 (0.790)	79.794 (6.396)
M.VII	4.175 (1.813)	5.650 (6.694)	0.842 (1.149)	89.333 (5.886)
M.VIII	3.541 (1.713)	4.146 (19.695)	1.452 (1.342)	90.861 (19.210)
M.IX (MAD)	4.423 (1.971)	3.633 (2.517)	0.581 (0.741)	91.363 (2.940)
M.X (IQR)	4.236 (1.911)	0.692 (1.026)	0.779 (0.853)	94.293 (2.172)

Table 14: Average values of the classification quality measures for the B1 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.935 (0.067)	0.573 (0.326)	0.854 (0.183)	0.618 (0.269)
M.II	0.849 (0.055)	0.241 (0.112)	0.854 (0.180)	0.362 (0.143)
M.III	0.874 (0.057)	0.302 (0.155)	0.923 (0.126)	0.432 (0.178)
M.III+BH	0.865 (0.048)	0.204 (0.123)	0.914 (0.158)	0.318 (0.157)
M.IV	0.976 (0.099)	0.927 (0.179)	0.849 (0.161)	0.863 (0.165)
M.V	0.913 (0.020)	0.320 (0.123)	0.684 (0.218)	0.416 (0.130)
M.VI	0.843 (0.073)	0.254 (0.132)	0.905 (0.142)	0.379 (0.164)
M.VII	0.935 (0.067)	0.573 (0.326)	0.854 (0.183)	0.618 (0.269)
M.VIII	0.944 (0.194)	0.956 (0.199)	0.732 (0.222)	0.782 (0.224)
M.IX (MAD)	0.958 (0.025)	0.575 (0.220)	0.887 (0.148)	0.670 (0.182)
M.X (IQR)	0.985 (0.013)	0.866 (0.186)	0.848 (0.170)	0.838 (0.150)

Table 15: Average values of TP/FP/FN/TN for the B1 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	42.555 (7.037)	20.530 (26.726)	7.267 (4.394)	929.648 (26.872)
M.II	43.144 (6.810)	106.976 (14.240)	6.678 (2.945)	843.202 (14.587)
M.III	45.568 (6.632)	107.984 (11.915)	4.254 (2.041)	842.194 (11.695)
M.III+BH	41.799 (6.682)	15.441 (10.715)	7.553 (4.048)	935.208 (11.434)
M.IV	41.328 (6.746)	2.442 (2.001)	8.494 (3.033)	947.736 (7.629)
M.V	38.828 (3.352)	61.246 (3.470)	10.994 (5.091)	888.932 (5.206)
M.VI	45.141 (6.657)	111.777 (22.730)	4.681 (2.189)	838.401 (22.588)
M.VII	42.536 (7.039)	20.519 (26.728)	7.286 (4.395)	929.659 (26.876)
M.VIII	32.653 (6.836)	0.028 (0.165)	17.169 (5.589)	950.150 (6.934)
M.IX (MAD)	44.017 (6.511)	31.264 (7.559)	5.805 (2.343)	918.914 (9.120)
M.X (IQR)	41.779 (6.412)	4.417 (2.464)	8.043 (2.813)	945.761 (7.081)

Table 16: Average values of the classification quality measures for the B1 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.972 (0.029)	0.708 (0.129)	0.855 (0.086)	0.767 (0.097)
M.II	0.886 (0.015)	0.288 (0.041)	0.866 (0.058)	0.431 (0.050)
M.III	0.888 (0.012)	0.298 (0.043)	0.915 (0.040)	0.447 (0.050)
M.III+BH	0.977 (0.010)	0.749 (0.121)	0.848 (0.077)	0.786 (0.076)
M.IV	0.989 (0.003)	0.947 (0.040)	0.829 (0.059)	0.882 (0.036)
M.V	0.928 (0.005)	0.388 (0.033)	0.787 (0.074)	0.518 (0.032)
M.VI	0.884 (0.023)	0.290 (0.044)	0.906 (0.043)	0.438 (0.053)
M.VII	0.972 (0.029)	0.708 (0.129)	0.854 (0.086)	0.767 (0.097)
M.VIII	0.983 (0.006)	0.999 (0.004)	0.655 (0.101)	0.787 (0.078)
M.IX (MAD)	0.963 (0.008)	0.587 (0.074)	0.883 (0.045)	0.703 (0.057)
M.X (IQR)	0.988 (0.004)	0.905 (0.051)	0.838 (0.053)	0.869 (0.038)

Table 17: Average values of TP/FP/FN/TN for the B2 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	1.875 (1.026)	18.974 (8.722)	3.492 (1.848)	75.659 (8.747)
M.II	1.984 (1.092)	24.014 (7.432)	3.300 (1.772)	70.702 (7.505)
M.III	1.991 (1.051)	23.098 (6.816)	3.333 (1.792)	71.578 (6.800)
M.III+BH	1.842 (0.950)	18.417 (6.495)	3.515 (1.891)	76.226 (6.551)
M.IV	2.419 (1.951)	37.724 (35.581)	2.864 (2.405)	56.993 (35.226)
M.V	1.502 (0.710)	8.978 (1.595)	4.056 (1.981)	85.464 (2.421)
M.VI	2.135 (1.188)	26.926 (7.840)	3.159 (1.761)	67.780 (7.841)
M.VII	1.875 (1.026)	18.975 (8.726)	3.492 (1.848)	75.658 (8.751)
M.VIII	4.516 (2.315)	85.081 (29.226)	0.487 (1.587)	9.916 (28.941)
M.IX (MAD)	1.352 (0.617)	4.566 (2.897)	4.221 (2.016)	89.861 (3.349)
M.X (IQR)	1.171 (0.435)	1.124 (1.489)	4.487 (2.034)	93.218 (2.597)

Table 18: Average values of the classification quality measures for the B2 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.775 (0.085)	0.106 (0.095)	0.378 (0.194)	0.149 (0.074)
M.II	0.727 (0.073)	0.079 (0.043)	0.402 (0.196)	0.127 (0.063)
M.III	0.736 (0.067)	0.084 (0.049)	0.404 (0.199)	0.133 (0.066)
M.III+BH	0.781 (0.064)	0.096 (0.051)	0.378 (0.194)	0.146 (0.068)
M.IV	0.594 (0.339)	0.135 (0.177)	0.504 (0.338)	0.145 (0.100)
M.V	0.870 (0.025)	0.146 (0.070)	0.307 (0.178)	0.188 (0.084)
M.VI	0.699 (0.076)	0.076 (0.043)	0.430 (0.211)	0.125 (0.064)
M.VII	0.775 (0.085)	0.106 (0.095)	0.378 (0.194)	0.149 (0.074)
M.VIII	0.144 (0.278)	0.146 (0.287)	0.918 (0.243)	0.119 (0.097)
M.IX (MAD)	0.912 (0.033)	0.288 (0.187)	0.280 (0.168)	0.248 (0.108)
M.X (IQR)	0.944 (0.025)	0.670 (0.306)	0.244 (0.152)	0.319 (0.121)

Table 19: Average values of TP/FP/FN/TN for the B2 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	11.406 (3.794)	130.556 (45.048)	38.416 (6.263)	819.622 (45.056)
M.II	13.461 (3.943)	182.192 (32.859)	36.361 (5.843)	767.986 (32.898)
M.III	15.135 (4.088)	208.045 (34.296)	34.687 (5.831)	742.133 (34.420)
M.III+BH	11.774 (3.721)	137.338 (38.982)	38.064 (6.168)	812.825 (39.052)
M.IV	16.530 (20.497)	281.080 (411.231)	33.309 (21.256)	669.081 (410.869)
M.V	9.014 (2.828)	91.065 (3.211)	40.808 (6.170)	859.113 (6.288)
M.VI	15.274 (4.162)	212.604 (40.617)	34.548 (5.954)	737.574 (40.558)
M.VII	11.405 (3.793)	130.539 (45.050)	38.417 (6.262)	819.639 (45.056)
M.VIII	22.209 (23.953)	414.843 (470.254)	27.597 (24.751)	535.351 (469.597)
M.IX (MAD)	6.659 (2.574)	40.449 (8.867)	43.166 (6.432)	909.726 (11.031)
M.X (IQR)	3.635 (1.817)	6.538 (3.254)	46.275 (6.661)	943.552 (7.688)

Table 20: Average values of the classification quality measures for the B2 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.831 (0.043)	0.084 (0.029)	0.229 (0.069)	0.120 (0.035)
M.II	0.781 (0.032)	0.069 (0.019)	0.270 (0.067)	0.110 (0.028)
M.III	0.757 (0.033)	0.069 (0.019)	0.303 (0.069)	0.111 (0.028)
M.III+BH	0.825 (0.038)	0.082 (0.028)	0.236 (0.067)	0.120 (0.035)
M.IV	0.686 (0.391)	0.460 (0.407)	0.335 (0.409)	0.100 (0.039)
M.V	0.868 (0.007)	0.090 (0.028)	0.181 (0.051)	0.120 (0.035)
M.VI	0.753 (0.039)	0.068 (0.018)	0.306 (0.072)	0.110 (0.027)
M.VII	0.831 (0.043)	0.084 (0.029)	0.229 (0.069)	0.120 (0.035)
M.VIII	0.558 (0.446)	0.557 (0.466)	0.453 (0.480)	0.073 (0.029)
M.IX (MAD)	0.916 (0.011)	0.143 (0.054)	0.134 (0.048)	0.136 (0.048)
M.X (IQR)	0.947 (0.007)	0.373 (0.164)	0.073 (0.035)	0.119 (0.054)

Table 21: Average values of TP/FP/FN/TN for the B3 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	2.916 (1.545)	15.816 (6.015)	2.180 (1.415)	79.088 (6.311)
M.II	3.091 (1.602)	23.029 (5.407)	1.996 (1.375)	71.884 (5.656)
M.III	3.127 (1.606)	22.253 (5.459)	1.953 (1.319)	72.667 (5.705)
M.III+BH	3.093 (1.559)	16.452 (5.642)	2.142 (1.376)	78.313 (5.906)
M.IV	2.519 (1.493)	30.175 (11.721)	2.668 (1.848)	64.638 (11.539)
M.V	2.337 (1.011)	7.742 (1.559)	2.785 (1.683)	87.136 (2.059)
M.VI	3.191 (1.645)	24.108 (5.782)	1.879 (1.324)	70.822 (5.989)
M.VII	2.917 (1.542)	15.818 (6.020)	2.181 (1.417)	79.084 (6.316)
M.VIII	4.493 (2.306)	84.988 (29.141)	0.536 (1.689)	9.983 (28.690)
M.IX (MAD)	1.965 (1.049)	11.848 (3.475)	3.364 (1.841)	82.823 (3.828)
M.X (IQR)	1.458 (0.722)	5.312 (2.313)	4.027 (2.034)	89.203 (3.015)

Table 22: Average values of the classification quality measures for the B3 scenario; $n = 100$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.820 (0.060)	0.165 (0.090)	0.588 (0.216)	0.245 (0.108)
M.II	0.750 (0.055)	0.120 (0.061)	0.623 (0.221)	0.196 (0.088)
M.III	0.758 (0.055)	0.125 (0.062)	0.630 (0.213)	0.203 (0.089)
M.III+BH	0.814 (0.056)	0.164 (0.081)	0.604 (0.210)	0.248 (0.106)
M.IV	0.672 (0.114)	0.089 (0.062)	0.517 (0.246)	0.141 (0.081)
M.V	0.895 (0.021)	0.234 (0.103)	0.494 (0.202)	0.303 (0.112)
M.VI	0.740 (0.058)	0.119 (0.060)	0.643 (0.216)	0.195 (0.088)
M.VII	0.820 (0.060)	0.165 (0.090)	0.588 (0.217)	0.245 (0.108)
M.VIII	0.145 (0.276)	0.116 (0.227)	0.916 (0.246)	0.114 (0.083)
M.IX (MAD)	0.848 (0.036)	0.146 (0.074)	0.401 (0.203)	0.203 (0.090)
M.X (IQR)	0.907 (0.029)	0.237 (0.129)	0.306 (0.185)	0.241 (0.092)

Table 23: Average values of TP/FP/FN/TN for the B3 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	28.213 (5.325)	145.644 (37.055)	21.609 (4.698)	804.534 (37.247)
M.II	29.639 (5.519)	206.000 (32.456)	20.183 (4.559)	744.178 (32.737)
M.III	30.791 (5.694)	224.188 (33.465)	19.031 (4.378)	725.990 (33.540)
M.III+BH	28.831 (5.452)	158.899 (39.545)	20.991 (4.645)	791.279 (39.718)
M.IV	27.074 (6.744)	317.026 (70.048)	22.748 (6.390)	633.152 (70.031)
M.V	26.385 (4.593)	73.599 (4.757)	23.437 (4.822)	876.579 (5.013)
M.VI	30.751 (5.581)	225.816 (33.453)	19.071 (4.381)	724.362 (33.569)
M.VII	28.212 (5.323)	145.633 (37.059)	21.610 (4.699)	804.545 (37.249)
M.VIII	22.324 (24.210)	419.550 (471.053)	28.366 (25.695)	529.761 (469.738)
M.IX (MAD)	15.304 (3.930)	115.037 (10.700)	34.518 (5.831)	835.141 (12.437)
M.X (IQR)	7.704 (2.762)	47.360 (7.112)	42.118 (6.254)	902.818 (9.778)

Table 24: Average values of the classification quality measures for the B3 scenario; $n = 1000$, 5% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.833 (0.036)	0.169 (0.043)	0.566 (0.074)	0.257 (0.052)
M.II	0.774 (0.032)	0.127 (0.026)	0.595 (0.074)	0.209 (0.037)
M.III	0.757 (0.033)	0.122 (0.025)	0.618 (0.073)	0.203 (0.036)
M.III+BH	0.820 (0.039)	0.159 (0.040)	0.579 (0.075)	0.247 (0.050)
M.IV	0.660 (0.067)	0.080 (0.018)	0.543 (0.112)	0.139 (0.028)
M.V	0.903 (0.007)	0.264 (0.046)	0.530 (0.067)	0.351 (0.051)
M.VI	0.755 (0.033)	0.121 (0.024)	0.617 (0.072)	0.202 (0.035)
M.VII	0.833 (0.036)	0.169 (0.043)	0.566 (0.074)	0.257 (0.052)
M.VIII	0.552 (0.446)	0.275 (0.314)	0.453 (0.485)	0.066 (0.031)
M.IX (MAD)	0.850 (0.012)	0.117 (0.028)	0.307 (0.067)	0.169 (0.038)
M.X (IQR)	0.911 (0.009)	0.140 (0.046)	0.154 (0.051)	0.145 (0.046)

Table 25: Average values of TP/FP/FN/TN for the A1 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	10.048 (2.977)	0.761 (1.398)	0.012 (0.383)	89.178 (2.951)
M.II	9.933 (3.065)	10.055 (1.612)	0.120 (0.926)	79.892 (2.731)
M.III	9.811 (2.821)	8.126 (1.536)	0.223 (0.910)	81.840 (2.684)
M.III+BH	6.000 (5.292)	11.333 (9.292)	0.000 (0.000)	82.667 (4.041)
M.IV	10.037 (2.971)	0.036 (0.302)	0.000 (0.000)	89.927 (2.983)
M.V	3.302 (0.983)	6.548 (1.387)	6.735 (2.913)	83.415 (3.056)
M.VI	9.929 (3.081)	10.377 (2.705)	0.113 (0.898)	79.581 (3.499)
M.VII	10.048 (2.977)	0.760 (1.398)	0.012 (0.383)	89.179 (2.951)
M.VIII	10.033 (2.972)	0.005 (0.071)	0.001 (0.032)	89.961 (2.972)
M.IX (MAD)	10.034 (2.973)	2.318 (2.057)	0.000 (0.000)	87.648 (3.020)
M.X (IQR)	10.034 (2.973)	0.323 (0.627)	0.000 (0.000)	89.643 (2.923)

Table 26: Average values of the classification quality measures for the A1 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.992 (0.015)	0.930 (0.106)	0.999 (0.029)	0.960 (0.078)
M.II	0.898 (0.022)	0.488 (0.101)	0.988 (0.085)	0.648 (0.105)
M.III	0.917 (0.015)	0.540 (0.100)	0.983 (0.068)	0.689 (0.088)
M.III+BH	0.887 (0.093)	0.413 (0.446)	1.000 (0.000)	0.499 (0.410)
M.IV	1.000 (0.003)	0.997 (0.025)	1.000 (0.000)	0.998 (0.014)
M.V	0.867 (0.032)	0.337 (0.098)	0.357 (0.152)	0.336 (0.098)
M.VI	0.895 (0.033)	0.485 (0.108)	0.988 (0.087)	0.645 (0.112)
M.VII	0.992 (0.015)	0.930 (0.106)	0.999 (0.029)	0.960 (0.078)
M.VIII	1.000 (0.001)	0.999 (0.008)	1.000 (0.002)	1.000 (0.004)
M.IX (MAD)	0.977 (0.021)	0.815 (0.151)	1.000 (0.000)	0.890 (0.102)
M.X (IQR)	0.997 (0.006)	0.967 (0.068)	1.000 (0.000)	0.982 (0.040)

Table 27: Average values of TP/FP/FN/TN for the A1 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	99.566 (9.984)	9.407 (8.101)	0.089 (2.814)	890.938 (11.978)
M.II	99.116 (9.836)	95.775 (4.739)	0.539 (2.426)	804.570 (9.360)
M.III	99.610 (9.505)	96.744 (3.858)	0.045 (0.627)	803.601 (9.035)
M.III+BH	64.000 (7.071)	20.000 (11.314)	26.500 (12.021)	889.500 (6.364)
M.IV	99.645 (9.488)	0.013 (0.113)	0.010 (0.100)	900.332 (9.488)
M.V	35.785 (1.770)	64.327 (1.966)	63.870 (8.903)	836.018 (8.969)
M.VI	99.600 (9.702)	98.476 (6.880)	0.055 (1.739)	801.869 (10.463)
M.VII	99.566 (9.984)	9.405 (8.102)	0.089 (2.814)	890.940 (11.977)
M.VIII	99.642 (9.485)	0.020 (0.140)	0.013 (0.122)	900.325 (9.487)
M.IX (MAD)	99.655 (9.489)	20.363 (5.811)	0.000 (0.000)	879.982 (9.503)
M.X (IQR)	99.655 (9.489)	2.178 (1.624)	0.000 (0.000)	898.167 (9.310)

Table 28: Average values of the classification quality measures for the A1 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.991 (0.011)	0.915 (0.040)	0.999 (0.031)	0.955 (0.034)
M.II	0.904 (0.006)	0.508 (0.031)	0.994 (0.026)	0.672 (0.030)
M.III	0.903 (0.004)	0.506 (0.029)	1.000 (0.007)	0.672 (0.025)
M.III+BH	0.954 (0.001)	0.771 (0.085)	0.711 (0.117)	0.732 (0.025)
M.IV	1.000 (0.000)	1.000 (0.001)	1.000 (0.001)	1.000 (0.001)
M.V	0.872 (0.009)	0.357 (0.017)	0.362 (0.032)	0.359 (0.018)
M.VI	0.901 (0.008)	0.502 (0.031)	0.999 (0.019)	0.668 (0.030)
M.VII	0.991 (0.011)	0.915 (0.040)	0.999 (0.031)	0.955 (0.034)
M.VIII	1.000 (0.000)	1.000 (0.001)	1.000 (0.001)	1.000 (0.001)
M.IX (MAD)	0.980 (0.006)	0.830 (0.046)	1.000 (0.000)	0.907 (0.028)
M.X (IQR)	0.998 (0.002)	0.978 (0.016)	1.000 (0.000)	0.989 (0.008)

Table 29: Average values of TP/FP/FN/TN for the A2 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	10.039 (2.973)	4.553 (3.934)	0.006 (0.078)	85.402 (3.786)
M.II	9.813 (3.086)	12.796 (3.804)	0.244 (1.148)	77.148 (3.765)
M.III	9.819 (2.887)	11.295 (3.895)	0.215 (0.975)	78.671 (3.753)
M.III+BH	7.521 (1.951)	8.829 (5.784)	0.150 (0.574)	83.500 (5.000)
M.IV	9.839 (2.954)	0.776 (4.501)	0.209 (0.471)	89.176 (5.280)
M.V	3.148 (1.000)	7.024 (1.430)	6.893 (2.854)	82.935 (3.047)
M.VI	9.870 (3.138)	13.954 (5.039)	0.164 (1.107)	76.012 (4.775)
M.VII	10.039 (2.973)	4.551 (3.935)	0.006 (0.078)	85.404 (3.785)
M.VIII	9.515 (3.363)	27.108 (40.992)	0.519 (1.455)	62.858 (41.639)
M.IX (MAD)	9.939 (2.909)	2.297 (1.966)	0.095 (0.338)	87.669 (2.986)
M.X (IQR)	9.333 (2.653)	0.345 (0.664)	0.701 (1.166)	89.621 (2.915)

Table 30: Average values of the classification quality measures for the A2 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.954 (0.039)	0.709 (0.185)	0.999 (0.008)	0.814 (0.145)
M.II	0.870 (0.042)	0.435 (0.116)	0.977 (0.104)	0.593 (0.125)
M.III	0.885 (0.037)	0.472 (0.122)	0.983 (0.076)	0.625 (0.117)
M.III+BH	0.910 (0.057)	0.498 (0.175)	0.986 (0.055)	0.641 (0.167)
M.IV	0.990 (0.045)	0.955 (0.109)	0.979 (0.049)	0.961 (0.077)
M.V	0.861 (0.031)	0.311 (0.097)	0.337 (0.138)	0.313 (0.093)
M.VI	0.859 (0.056)	0.421 (0.122)	0.983 (0.106)	0.580 (0.135)
M.VII	0.954 (0.039)	0.709 (0.185)	0.999 (0.008)	0.814 (0.145)
M.VIII	0.724 (0.407)	0.726 (0.408)	0.942 (0.161)	0.713 (0.365)
M.IX (MAD)	0.976 (0.019)	0.815 (0.149)	0.992 (0.029)	0.886 (0.099)
M.X (IQR)	0.990 (0.012)	0.964 (0.072)	0.939 (0.094)	0.946 (0.065)

Table 31: Average values of TP/FP/FN/TN for the A2 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	99.566 (9.504)	36.934 (11.061)	0.080 (0.275)	863.420 (13.535)
M.II	96.912 (11.806)	113.244 (13.333)	2.743 (7.437)	787.101 (14.815)
M.III	99.229 (10.021)	120.965 (13.205)	0.426 (3.606)	779.380 (14.783)
M.III+BH	35.760 (25.548)	18.204 (13.008)	63.189 (29.836)	882.847 (13.300)
M.IV	97.662 (9.497)	1.635 (1.558)	1.993 (1.494)	898.710 (9.725)
M.V	33.153 (2.226)	66.894 (2.387)	66.502 (8.923)	833.451 (8.987)
M.VI	99.010 (11.311)	124.708 (18.473)	0.645 (6.322)	775.637 (19.678)
M.VII	99.565 (9.506)	36.877 (11.097)	0.081 (0.276)	863.477 (13.585)
M.VIII	87.537 (23.080)	128.339 (314.882)	12.118 (21.691)	772.006 (314.345)
M.IX (MAD)	99.061 (9.366)	20.267 (5.966)	0.594 (0.800)	880.078 (9.533)
M.X (IQR)	95.027 (8.662)	2.242 (1.759)	4.628 (2.680)	898.103 (9.339)

Table 32: Average values of the classification quality measures for the A2 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.963 (0.011)	0.732 (0.063)	0.999 (0.003)	0.844 (0.043)
M.II	0.884 (0.017)	0.461 (0.046)	0.973 (0.074)	0.625 (0.053)
M.III	0.879 (0.011)	0.452 (0.040)	0.996 (0.035)	0.620 (0.033)
M.III+BH	0.919 (0.019)	0.654 (0.071)	0.373 (0.278)	0.423 (0.214)
M.IV	0.996 (0.002)	0.984 (0.015)	0.980 (0.015)	0.982 (0.010)
M.V	0.867 (0.009)	0.331 (0.022)	0.335 (0.032)	0.332 (0.022)
M.VI	0.875 (0.024)	0.444 (0.047)	0.994 (0.063)	0.613 (0.054)
M.VII	0.963 (0.011)	0.733 (0.063)	0.999 (0.003)	0.844 (0.043)
M.VIII	0.860 (0.311)	0.869 (0.314)	0.879 (0.216)	0.795 (0.316)
M.IX (MAD)	0.979 (0.006)	0.830 (0.047)	0.994 (0.008)	0.904 (0.028)
M.X (IQR)	0.993 (0.003)	0.977 (0.018)	0.954 (0.025)	0.965 (0.014)

Table 33: Average values of TP/FP/FN/TN for the A3 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	3.554 (1.988)	16.962 (7.335)	6.547 (2.953)	72.937 (7.460)
M.II	3.697 (1.978)	21.975 (6.433)	6.398 (2.835)	67.930 (6.546)
M.III	3.955 (1.867)	20.989 (6.290)	6.136 (2.686)	68.920 (6.471)
M.III+BH	3.360 (1.638)	16.236 (5.606)	6.706 (2.781)	73.698 (5.823)
M.IV	4.718 (3.770)	33.328 (32.507)	5.470 (4.171)	56.484 (32.359)
M.V	1.942 (0.901)	8.565 (1.567)	8.284 (2.837)	81.209 (3.089)
M.VI	4.251 (2.073)	24.773 (6.887)	5.825 (2.768)	65.150 (6.976)
M.VII	3.554 (1.988)	16.962 (7.335)	6.547 (2.953)	72.937 (7.460)
M.VIII	9.821 (3.231)	87.761 (14.193)	0.214 (1.450)	2.205 (13.946)
M.IX (MAD)	1.754 (0.912)	3.817 (2.569)	8.577 (2.968)	85.852 (3.606)
M.X (IQR)	1.178 (0.434)	1.010 (1.153)	8.973 (2.927)	88.839 (3.059)

Table 34: Average values of the classification quality measures for the A3 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.765 (0.069)	0.181 (0.091)	0.364 (0.193)	0.229 (0.102)
M.II	0.716 (0.061)	0.145 (0.069)	0.377 (0.189)	0.202 (0.090)
M.III	0.729 (0.061)	0.163 (0.075)	0.403 (0.173)	0.224 (0.090)
M.III+BH	0.771 (0.055)	0.177 (0.082)	0.347 (0.160)	0.224 (0.092)
M.IV	0.612 (0.298)	0.213 (0.195)	0.481 (0.348)	0.217 (0.125)
M.V	0.832 (0.032)	0.187 (0.088)	0.200 (0.094)	0.187 (0.080)
M.VI	0.694 (0.066)	0.148 (0.066)	0.433 (0.194)	0.214 (0.088)
M.VII	0.765 (0.069)	0.181 (0.091)	0.364 (0.193)	0.229 (0.102)
M.VIII	0.120 (0.129)	0.122 (0.140)	0.979 (0.135)	0.182 (0.053)
M.IX (MAD)	0.876 (0.035)	0.374 (0.213)	0.182 (0.103)	0.222 (0.100)
M.X (IQR)	0.900 (0.030)	0.670 (0.295)	0.128 (0.065)	0.200 (0.073)

Table 35: Average values of TP/FP/FN/TN for the A3 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	27.820 (8.780)	121.929 (40.491)	71.835 (10.959)	778.416 (41.254)
M.II	29.799 (8.622)	172.024 (32.346)	69.856 (10.970)	728.321 (33.003)
M.III	37.376 (7.102)	192.026 (30.129)	62.279 (8.641)	708.319 (30.867)
M.III+BH	28.538 (7.247)	128.214 (32.797)	71.115 (9.708)	772.132 (33.735)
M.IV	55.213 (40.851)	445.677 (379.668)	44.535 (41.085)	454.575 (379.447)
M.V	17.205 (3.423)	82.932 (3.641)	82.450 (8.688)	817.413 (8.740)
M.VI	36.642 (8.318)	198.273 (36.613)	63.013 (9.762)	702.072 (37.200)
M.VII	27.818 (8.781)	121.923 (40.491)	71.837 (10.961)	778.422 (41.255)
M.VIII	92.353 (27.313)	831.082 (236.053)	7.362 (25.533)	69.203 (236.137)
M.IX (MAD)	11.675 (3.607)	33.474 (7.625)	87.980 (9.286)	866.871 (11.549)
M.X (IQR)	3.108 (1.656)	5.057 (2.708)	96.751 (9.479)	895.084 (9.609)

Table 36: Average values of the classification quality measures for the A3 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.806 (0.036)	0.189 (0.041)	0.279 (0.085)	0.221 (0.045)
M.II	0.758 (0.029)	0.147 (0.031)	0.299 (0.083)	0.196 (0.041)
M.III	0.746 (0.027)	0.164 (0.026)	0.375 (0.062)	0.227 (0.032)
M.III+BH	0.801 (0.029)	0.185 (0.035)	0.287 (0.068)	0.221 (0.037)
M.IV	0.510 (0.344)	0.203 (0.196)	0.555 (0.406)	0.182 (0.101)
M.V	0.835 (0.009)	0.172 (0.034)	0.173 (0.032)	0.172 (0.032)
M.VI	0.739 (0.033)	0.156 (0.026)	0.368 (0.076)	0.218 (0.034)
M.VII	0.806 (0.036)	0.189 (0.041)	0.279 (0.085)	0.221 (0.045)
M.VIII	0.162 (0.211)	0.157 (0.214)	0.925 (0.257)	0.171 (0.045)
M.IX (MAD)	0.879 (0.011)	0.260 (0.068)	0.117 (0.035)	0.160 (0.043)
M.X (IQR)	0.898 (0.010)	0.396 (0.170)	0.031 (0.017)	0.057 (0.029)

Table 37: Average values of TP/FP/FN/TN for the B1 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	7.600 (2.775)	1.892 (4.136)	2.438 (2.243)	88.070 (4.488)
M.II	7.856 (2.679)	10.684 (2.968)	2.178 (1.827)	79.282 (3.167)
M.III	8.910 (2.718)	7.626 (3.164)	1.124 (1.129)	82.340 (2.905)
M.III+BH	6.075 (2.336)	8.900 (3.781)	0.725 (1.012)	84.300 (2.747)
M.IV	8.312 (2.813)	0.584 (1.747)	1.721 (1.348)	89.383 (3.497)
M.V	4.433 (1.372)	5.492 (1.561)	5.601 (2.655)	84.474 (2.841)
M.VI	8.692 (2.783)	10.235 (4.202)	1.342 (1.456)	79.731 (4.065)
M.VII	7.600 (2.775)	1.890 (4.136)	2.438 (2.243)	88.072 (4.488)
M.VIII	6.738 (2.573)	0.444 (6.216)	3.296 (2.232)	89.522 (7.005)
M.IX (MAD)	8.692 (2.708)	2.557 (2.120)	1.342 (1.129)	87.409 (3.181)
M.X (IQR)	8.215 (2.563)	0.410 (0.761)	1.819 (1.399)	89.556 (2.935)

Table 38: Average values of the classification quality measures for the B1 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.957 (0.052)	0.850 (0.228)	0.768 (0.193)	0.786 (0.195)
M.II	0.871 (0.035)	0.422 (0.117)	0.790 (0.163)	0.541 (0.126)
M.III	0.912 (0.030)	0.547 (0.153)	0.892 (0.104)	0.662 (0.122)
M.III+BH	0.904 (0.035)	0.420 (0.181)	0.908 (0.132)	0.549 (0.168)
M.IV	0.977 (0.022)	0.949 (0.095)	0.827 (0.136)	0.874 (0.097)
M.V	0.889 (0.030)	0.448 (0.132)	0.467 (0.159)	0.444 (0.118)
M.VI	0.884 (0.046)	0.466 (0.133)	0.871 (0.130)	0.596 (0.133)
M.VII	0.957 (0.052)	0.850 (0.228)	0.768 (0.193)	0.786 (0.195)
M.VIII	0.963 (0.064)	0.995 (0.063)	0.677 (0.184)	0.788 (0.153)
M.IX (MAD)	0.961 (0.023)	0.781 (0.163)	0.868 (0.110)	0.810 (0.114)
M.X (IQR)	0.978 (0.015)	0.951 (0.090)	0.823 (0.127)	0.875 (0.090)

Table 39: Average values of TP/FP/FN/TN for the B1 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	78.712 (9.727)	9.679 (14.602)	20.943 (7.066)	890.666 (16.545)
M.II	86.061 (9.569)	90.746 (8.454)	13.594 (4.583)	809.599 (10.261)
M.III	90.430 (9.081)	86.871 (9.033)	9.225 (3.003)	813.474 (10.624)
M.III+BH	76.302 (10.732)	13.043 (8.086)	17.554 (8.461)	893.101 (11.320)
M.IV	84.368 (9.148)	5.604 (3.048)	15.287 (4.083)	894.741 (10.503)
M.V	48.188 (3.103)	51.817 (3.328)	51.467 (8.576)	848.528 (8.689)
M.VI	89.493 (9.459)	90.112 (13.400)	10.162 (3.954)	810.233 (14.127)
M.VII	78.703 (9.721)	9.638 (14.604)	20.952 (7.068)	890.707 (16.553)
M.VIII	64.239 (11.596)	0.023 (0.150)	35.416 (10.459)	900.322 (9.488)
M.IX (MAD)	86.747 (8.717)	21.529 (6.227)	12.908 (3.659)	878.816 (9.700)
M.X (IQR)	81.733 (8.308)	2.415 (1.756)	17.922 (4.352)	897.930 (9.382)

Table 40: Average values of the classification quality measures for the B1 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.969 (0.018)	0.898 (0.067)	0.790 (0.070)	0.838 (0.064)
M.II	0.896 (0.010)	0.486 (0.040)	0.863 (0.046)	0.621 (0.040)
M.III	0.904 (0.009)	0.510 (0.046)	0.907 (0.029)	0.652 (0.037)
M.III+BH	0.969 (0.008)	0.861 (0.071)	0.814 (0.086)	0.831 (0.048)
M.IV	0.979 (0.005)	0.939 (0.030)	0.846 (0.039)	0.889 (0.024)
M.V	0.897 (0.009)	0.482 (0.031)	0.487 (0.044)	0.483 (0.029)
M.VI	0.900 (0.015)	0.499 (0.043)	0.898 (0.040)	0.640 (0.042)
M.VII	0.969 (0.018)	0.898 (0.067)	0.790 (0.070)	0.838 (0.064)
M.VIII	0.965 (0.010)	1.000 (0.002)	0.645 (0.099)	0.779 (0.090)
M.IX (MAD)	0.966 (0.007)	0.802 (0.052)	0.871 (0.034)	0.834 (0.033)
M.X (IQR)	0.980 (0.004)	0.971 (0.021)	0.821 (0.038)	0.889 (0.024)

Table 41: Average values of TP/FP/FN/TN for the B2 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	2.886 (1.528)	16.973 (8.148)	7.351 (2.655)	72.790 (8.275)
M.II	3.354 (1.717)	21.937 (6.847)	6.803 (2.527)	67.906 (7.029)
M.III	3.285 (1.652)	21.109 (6.478)	6.901 (2.540)	68.706 (6.644)
M.III+BH	2.857 (1.481)	16.666 (5.842)	7.408 (2.633)	73.069 (6.117)
M.IV	3.895 (3.613)	29.802 (32.605)	6.325 (4.136)	59.979 (32.399)
M.V	1.998 (0.993)	8.378 (1.666)	8.383 (2.758)	81.241 (3.071)
M.VI	3.591 (1.739)	24.692 (7.441)	6.586 (2.497)	65.131 (7.534)
M.VII	2.886 (1.528)	16.973 (8.148)	7.351 (2.655)	72.790 (8.275)
M.VIII	8.073 (4.430)	71.540 (36.522)	1.967 (4.084)	18.420 (36.148)
M.IX (MAD)	1.856 (1.005)	4.247 (2.775)	8.666 (2.751)	85.231 (3.776)
M.X (IQR)	1.463 (0.725)	0.932 (1.281)	9.134 (2.878)	88.471 (3.225)

Table 42: Average values of the classification quality measures for the B2 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.757 (0.079)	0.165 (0.111)	0.289 (0.140)	0.194 (0.089)
M.II	0.713 (0.067)	0.137 (0.070)	0.334 (0.150)	0.188 (0.085)
M.III	0.720 (0.064)	0.141 (0.072)	0.328 (0.146)	0.189 (0.083)
M.III+BH	0.759 (0.059)	0.153 (0.079)	0.284 (0.134)	0.191 (0.087)
M.IV	0.639 (0.293)	0.235 (0.254)	0.394 (0.334)	0.187 (0.095)
M.V	0.832 (0.032)	0.195 (0.098)	0.201 (0.100)	0.192 (0.090)
M.VI	0.687 (0.072)	0.131 (0.062)	0.358 (0.151)	0.186 (0.078)
M.VII	0.757 (0.079)	0.165 (0.111)	0.289 (0.140)	0.194 (0.089)
M.VIII	0.265 (0.327)	0.280 (0.360)	0.817 (0.360)	0.182 (0.053)
M.IX (MAD)	0.871 (0.037)	0.362 (0.217)	0.184 (0.101)	0.224 (0.104)
M.X (IQR)	0.899 (0.031)	0.731 (0.285)	0.147 (0.078)	0.231 (0.099)

Table 43: Average values of TP/FP/FN/TN for the B2 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	21.437 (5.467)	108.997 (35.951)	78.218 (8.939)	791.348 (36.617)
M.II	26.031 (5.643)	162.086 (28.042)	73.624 (8.461)	738.259 (28.712)
M.III	29.300 (5.722)	185.726 (30.295)	70.355 (8.263)	714.619 (30.844)
M.III+BH	22.215 (5.518)	116.246 (35.145)	77.439 (8.865)	784.099 (35.875)
M.IV	15.588 (28.998)	105.733 (279.481)	84.160 (30.861)	794.520 (278.718)
M.V	17.713 (3.879)	82.319 (4.120)	81.942 (8.539)	818.026 (8.601)
M.VI	29.312 (6.017)	188.125 (32.431)	70.343 (8.495)	712.220 (32.972)
M.VII	21.437 (5.467)	108.975 (35.971)	78.218 (8.939)	791.370 (36.642)
M.VIII	22.604 (39.637)	193.052 (368.330)	77.047 (40.863)	707.296 (367.765)
M.IX (MAD)	13.288 (3.691)	37.287 (8.420)	86.367 (8.845)	863.058 (12.336)
M.X (IQR)	7.151 (2.729)	5.931 (3.069)	92.504 (9.147)	894.414 (9.900)

Table 44: Average values of the classification quality measures for the B2 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.813 (0.034)	0.171 (0.044)	0.215 (0.051)	0.186 (0.036)
M.II	0.764 (0.026)	0.139 (0.027)	0.261 (0.050)	0.180 (0.031)
M.III	0.744 (0.029)	0.138 (0.026)	0.294 (0.050)	0.186 (0.030)
M.III+BH	0.806 (0.033)	0.167 (0.042)	0.223 (0.051)	0.187 (0.035)
M.IV	0.810 (0.250)	0.729 (0.330)	0.159 (0.296)	0.102 (0.051)
M.V	0.836 (0.009)	0.177 (0.039)	0.178 (0.035)	0.177 (0.036)
M.VI	0.742 (0.030)	0.136 (0.026)	0.294 (0.053)	0.185 (0.030)
M.VII	0.813 (0.034)	0.171 (0.044)	0.215 (0.051)	0.186 (0.036)
M.VIII	0.730 (0.328)	0.787 (0.373)	0.229 (0.400)	0.069 (0.061)
M.IX (MAD)	0.876 (0.012)	0.266 (0.068)	0.133 (0.035)	0.176 (0.042)
M.X (IQR)	0.902 (0.009)	0.557 (0.156)	0.072 (0.026)	0.126 (0.043)

Table 45: Average values of TP/FP/FN/TN for the B3 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	5.571 (2.258)	15.283 (5.876)	4.478 (2.232)	74.668 (6.201)
M.II	5.908 (2.182)	22.016 (5.387)	4.141 (2.151)	67.935 (5.651)
M.III	6.011 (2.245)	21.277 (5.409)	4.038 (2.164)	68.674 (5.711)
M.III+BH	5.718 (2.254)	15.788 (5.643)	4.329 (2.029)	74.165 (6.258)
M.IV	4.351 (2.461)	25.695 (12.445)	5.718 (2.971)	64.236 (12.211)
M.V	3.429 (1.171)	6.549 (1.662)	6.620 (2.715)	83.402 (3.036)
M.VI	6.147 (2.350)	23.027 (5.670)	3.902 (2.060)	66.924 (5.779)
M.VII	5.571 (2.258)	15.283 (5.876)	4.478 (2.232)	74.668 (6.201)
M.VIII	8.345 (4.496)	71.373 (36.026)	1.918 (3.980)	18.364 (35.922)
M.IX (MAD)	3.312 (1.823)	11.619 (3.573)	6.875 (2.689)	78.193 (4.134)
M.X (IQR)	2.208 (1.343)	5.056 (2.131)	8.319 (2.693)	84.417 (3.453)

Table 46: Average values of the classification quality measures for the B3 scenario; $n = 100$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.802 (0.057)	0.279 (0.109)	0.560 (0.168)	0.359 (0.112)
M.II	0.738 (0.054)	0.215 (0.076)	0.596 (0.166)	0.309 (0.093)
M.III	0.747 (0.054)	0.224 (0.079)	0.606 (0.165)	0.319 (0.096)
M.III+BH	0.799 (0.057)	0.274 (0.101)	0.571 (0.156)	0.360 (0.109)
M.IV	0.686 (0.118)	0.176 (0.137)	0.443 (0.223)	0.224 (0.113)
M.V	0.868 (0.031)	0.347 (0.121)	0.358 (0.128)	0.343 (0.107)
M.VI	0.731 (0.056)	0.214 (0.079)	0.617 (0.162)	0.311 (0.097)
M.VII	0.802 (0.057)	0.279 (0.109)	0.560 (0.168)	0.359 (0.112)
M.VIII	0.267 (0.323)	0.211 (0.266)	0.816 (0.359)	0.185 (0.050)
M.IX (MAD)	0.815 (0.037)	0.222 (0.105)	0.331 (0.162)	0.255 (0.111)
M.X (IQR)	0.866 (0.032)	0.313 (0.161)	0.213 (0.120)	0.240 (0.111)

Table 47: Average values of TP/FP/FN/TN for the B3 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	TP	FP	FN	TN
M.I	55.837 (7.728)	139.974 (34.733)	43.818 (7.015)	760.371 (35.948)
M.II	59.281 (7.707)	196.481 (30.577)	40.374 (6.633)	703.864 (31.507)
M.III	61.660 (7.966)	213.083 (31.471)	37.995 (6.423)	687.262 (32.481)
M.III+BH	57.560 (7.804)	153.627 (37.043)	42.095 (6.830)	746.718 (38.219)
M.IV	53.522 (11.957)	285.060 (74.519)	46.133 (12.018)	615.285 (74.225)
M.V	38.370 (2.925)	61.738 (3.048)	61.285 (8.741)	838.607 (8.807)
M.VI	61.631 (7.997)	214.650 (31.769)	38.024 (6.403)	685.695 (32.762)
M.VII	55.836 (7.724)	139.971 (34.745)	43.819 (7.003)	760.374 (35.959)
M.VIII	28.113 (43.794)	247.420 (400.179)	72.263 (45.025)	652.204 (399.517)
M.IX (MAD)	30.272 (5.529)	111.431 (10.524)	69.383 (8.108)	788.914 (13.600)
M.X (IQR)	15.543 (4.076)	46.186 (7.024)	84.112 (8.681)	854.159 (11.594)

Table 48: Average values of the classification quality measures for the B3 scenario; $n = 1000$, 10% of outliers, 1000 replications.

Method	Accuracy	Precision	Recall	F1
M.I	0.816 (0.033)	0.292 (0.051)	0.560 (0.056)	0.381 (0.046)
M.II	0.763 (0.029)	0.234 (0.033)	0.595 (0.054)	0.334 (0.037)
M.III	0.749 (0.030)	0.226 (0.031)	0.619 (0.054)	0.330 (0.035)
M.III+BH	0.804 (0.035)	0.279 (0.049)	0.578 (0.056)	0.373 (0.045)
M.IV	0.669 (0.068)	0.163 (0.032)	0.537 (0.110)	0.245 (0.037)
M.V	0.819 (0.012)	0.214 (0.034)	0.304 (0.047)	0.250 (0.037)
M.VI	0.877 (0.009)	0.383 (0.029)	0.387 (0.037)	0.384 (0.027)
M.VII	0.747 (0.031)	0.225 (0.031)	0.618 (0.054)	0.329 (0.036)
M.VIII	0.816 (0.033)	0.292 (0.051)	0.560 (0.056)	0.381 (0.046)
M.IX (MAD)	0.680 (0.356)	0.380 (0.325)	0.283 (0.439)	0.070 (0.070)
M.X (IQR)	0.870 (0.011)	0.251 (0.055)	0.156 (0.038)	0.191 (0.042)

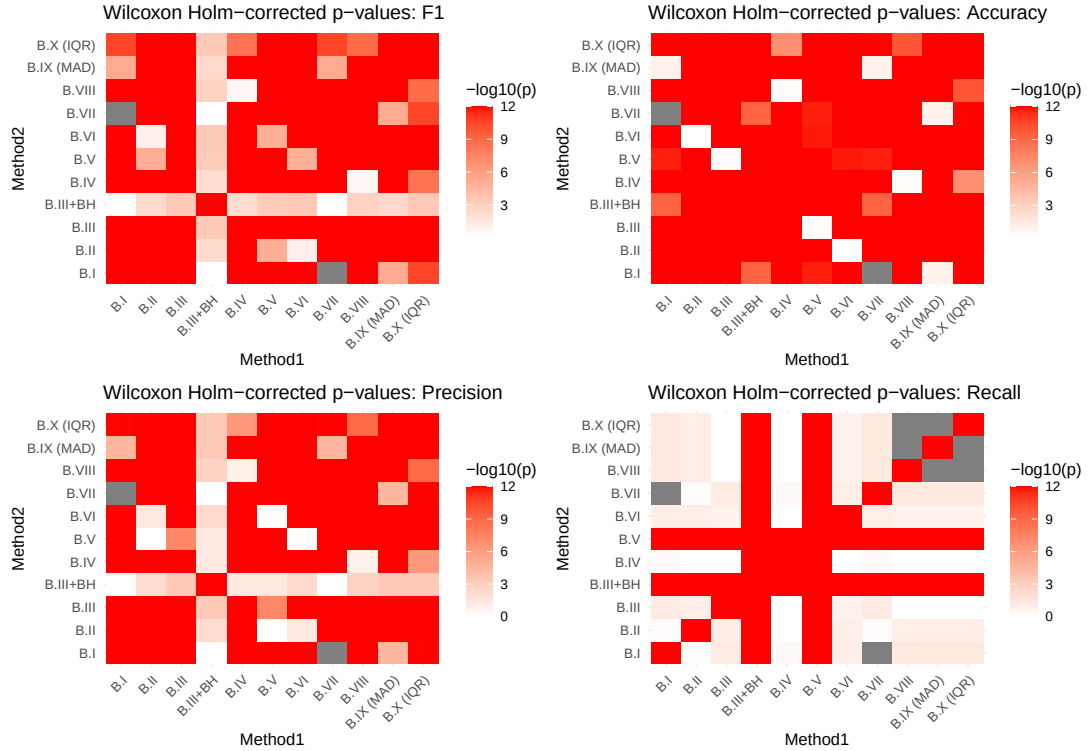


Figure 1: Adjusted p-values for Wilcoxon-Holm for A1, $n = 100$, $p = 0.05$.

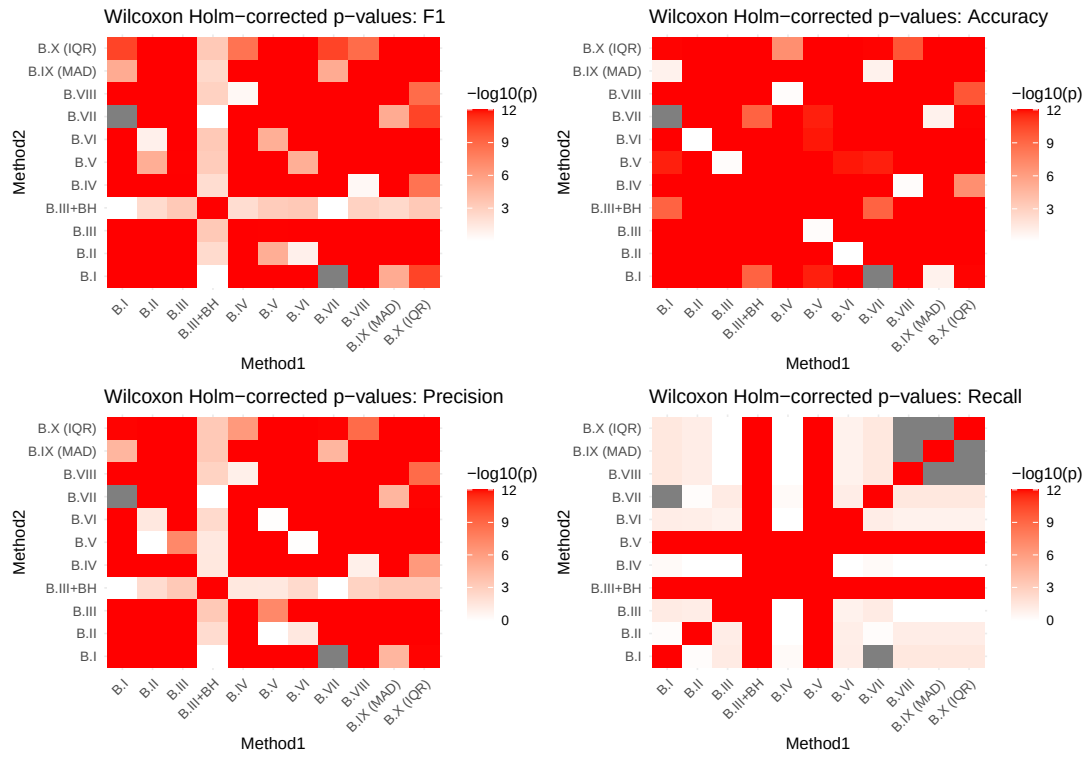


Figure 2: Adjusted p-values for Wilcoxon-Holm for A1, $n = 1000$, $p = 0.05$.

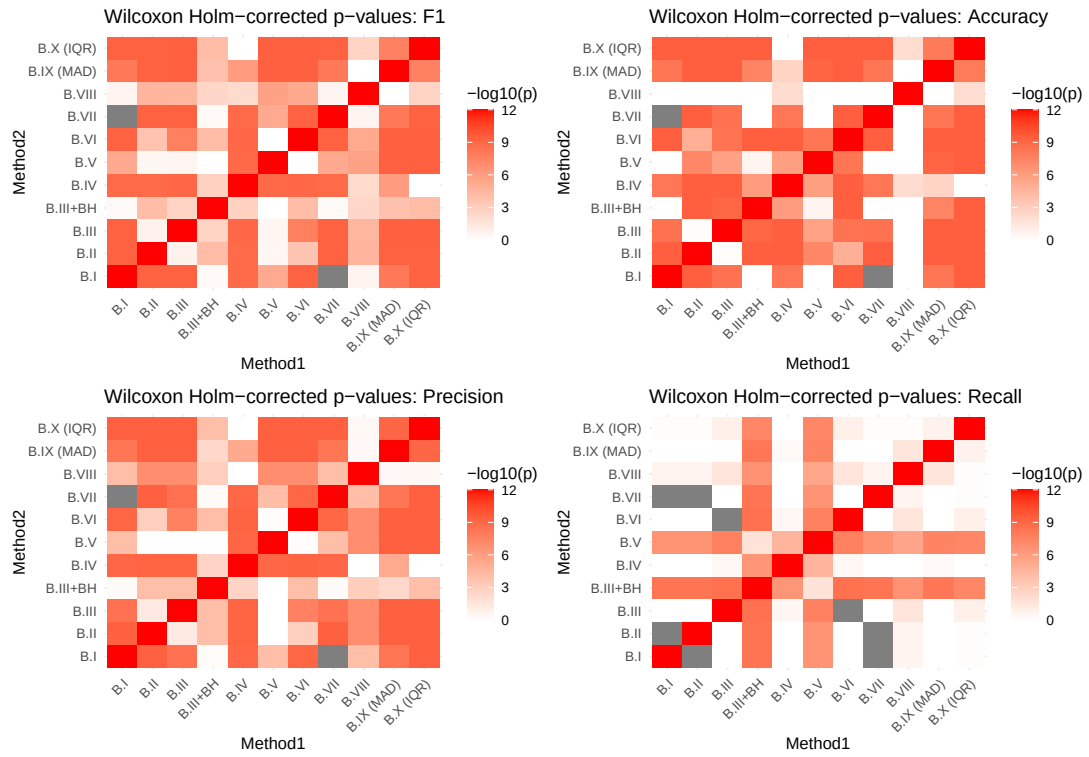


Figure 3: Adjusted p-values for Wilcoxon-Holm for A2, $n = 100$, $p = 0.05$.

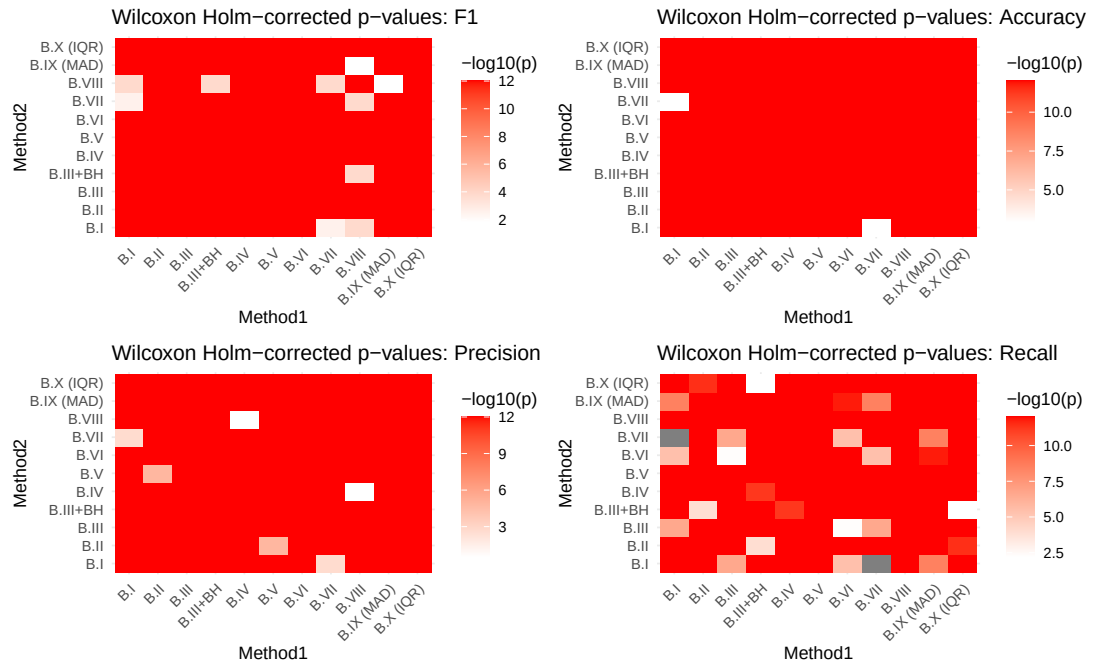


Figure 4: Adjusted p-values for Wilcoxon-Holm for A2, $n = 1000$, $p = 0.05$.

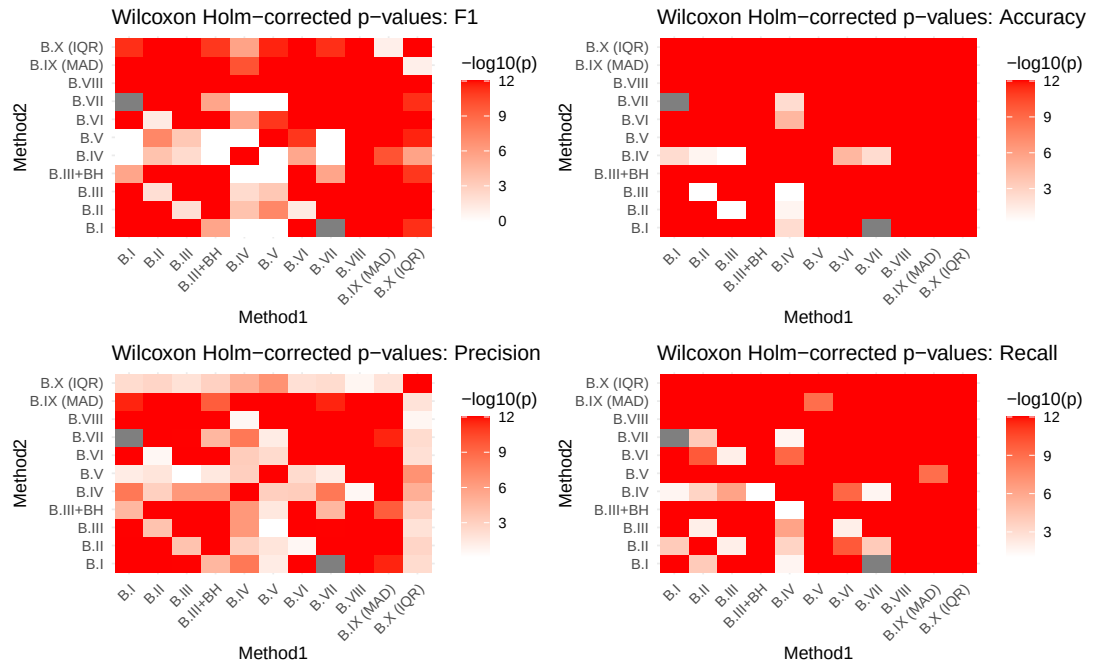


Figure 5: Adjusted p-values for Wilcoxon-Holm for A3, $n = 100$, $p = 0.05$.

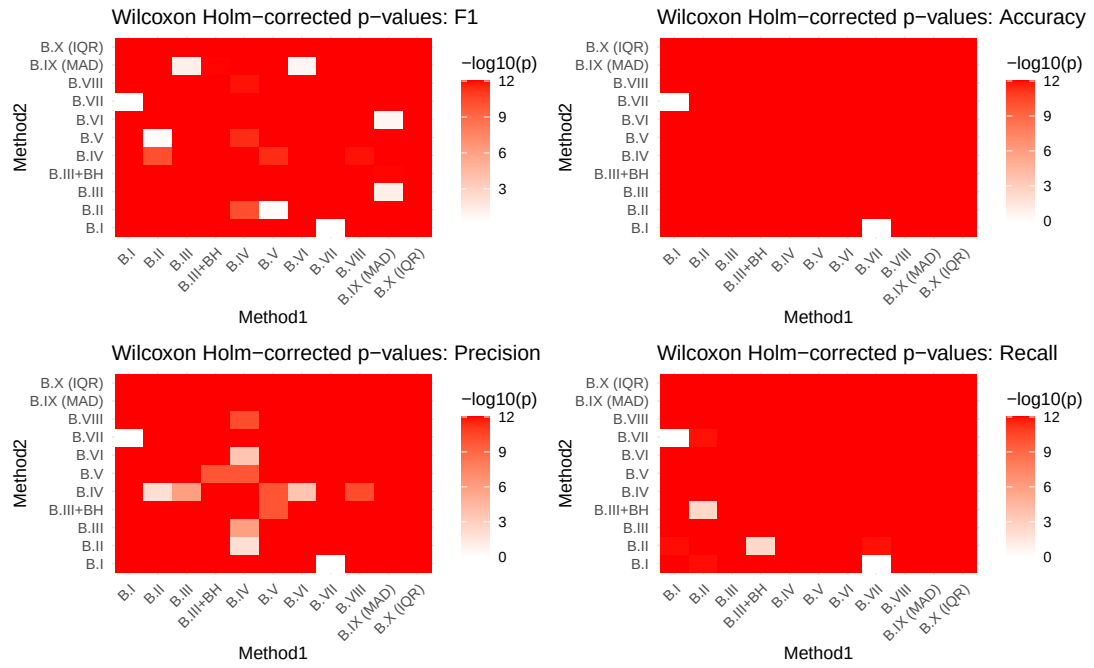


Figure 6: Adjusted p-values for Wilcoxon-Holm for A3, $n = 1000$, $p = 0.05$.

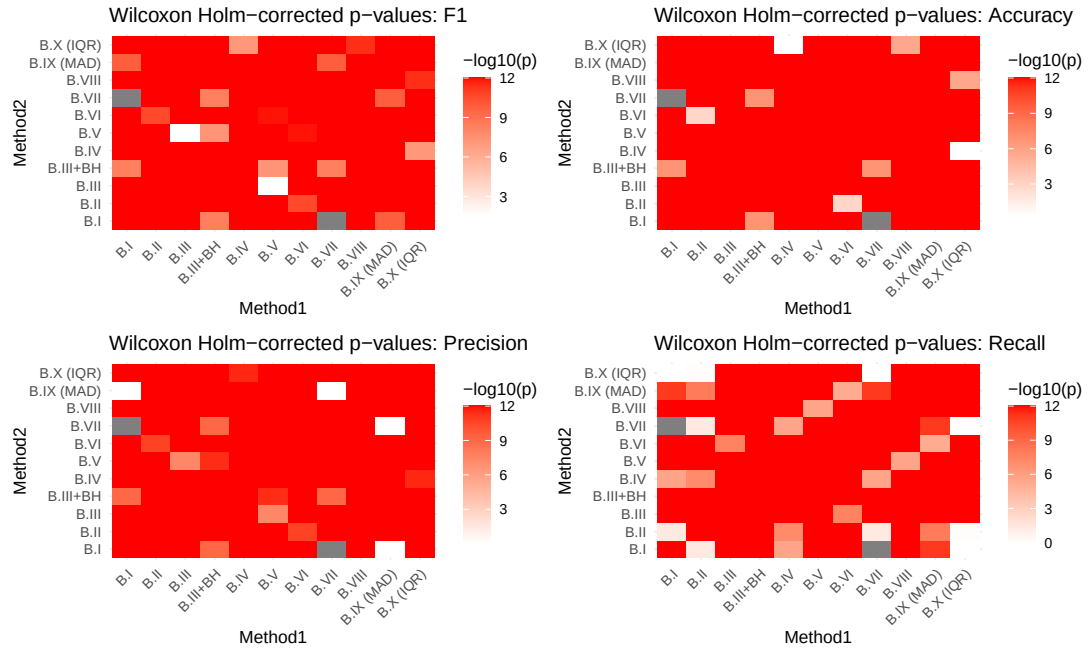


Figure 7: Adjusted p-values for Wilcoxon-Holm for B1, $n = 100$, $p = 0.05$.

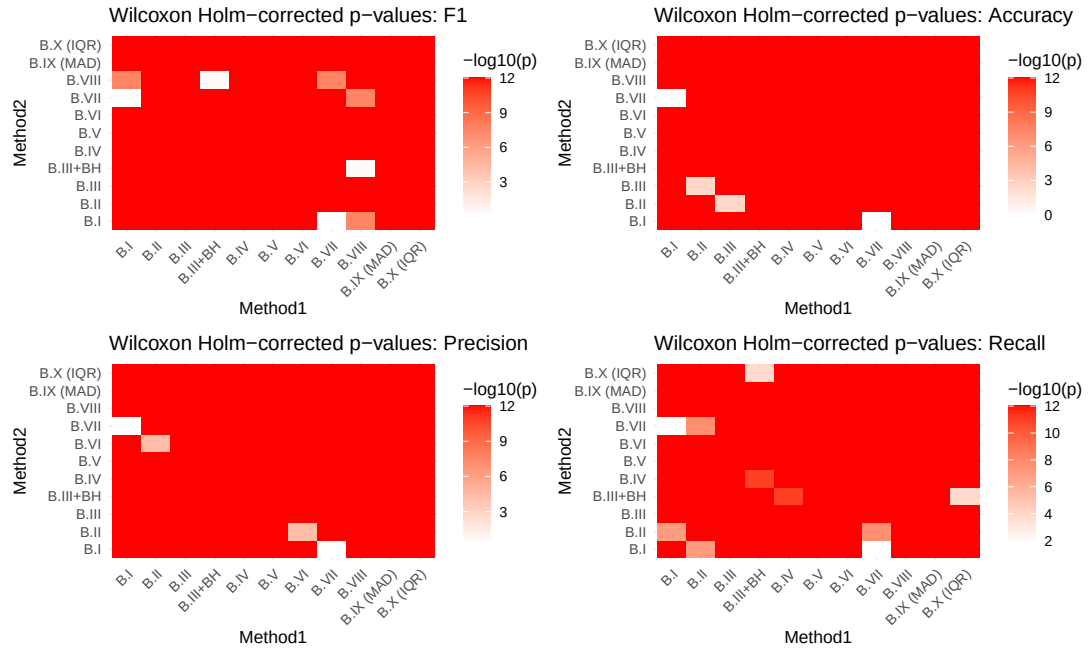


Figure 8: Adjusted p-values for Wilcoxon-Holm for B1, $n = 1000$, $p = 0.05$.

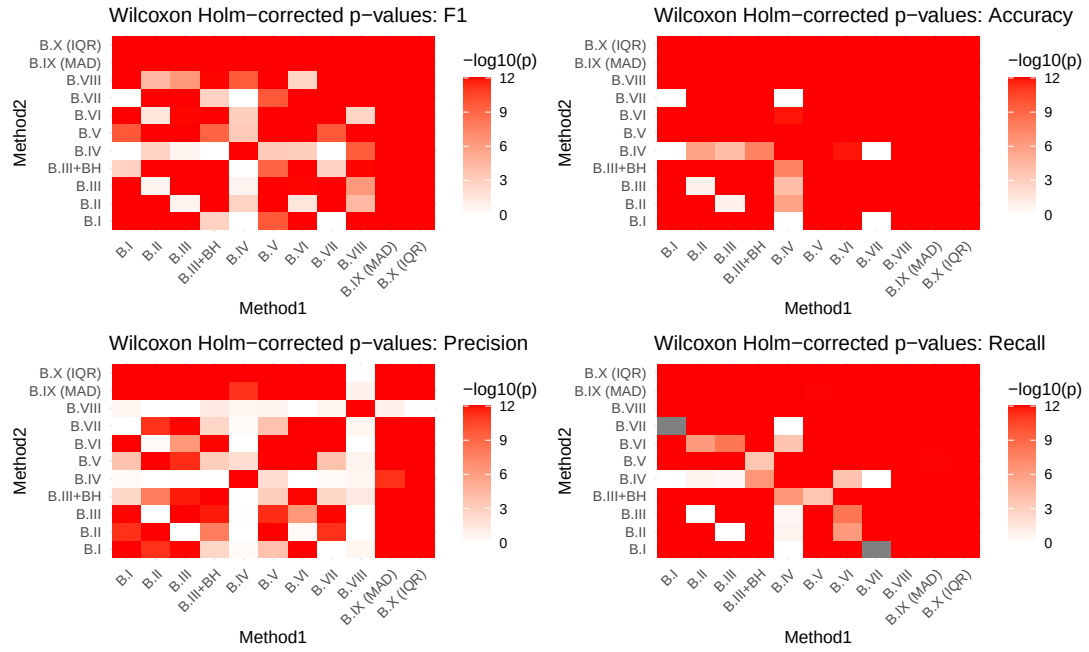


Figure 9: Adjusted p-values for Wilcoxon-Holm for B2, $n = 100$, $p = 0.05$.

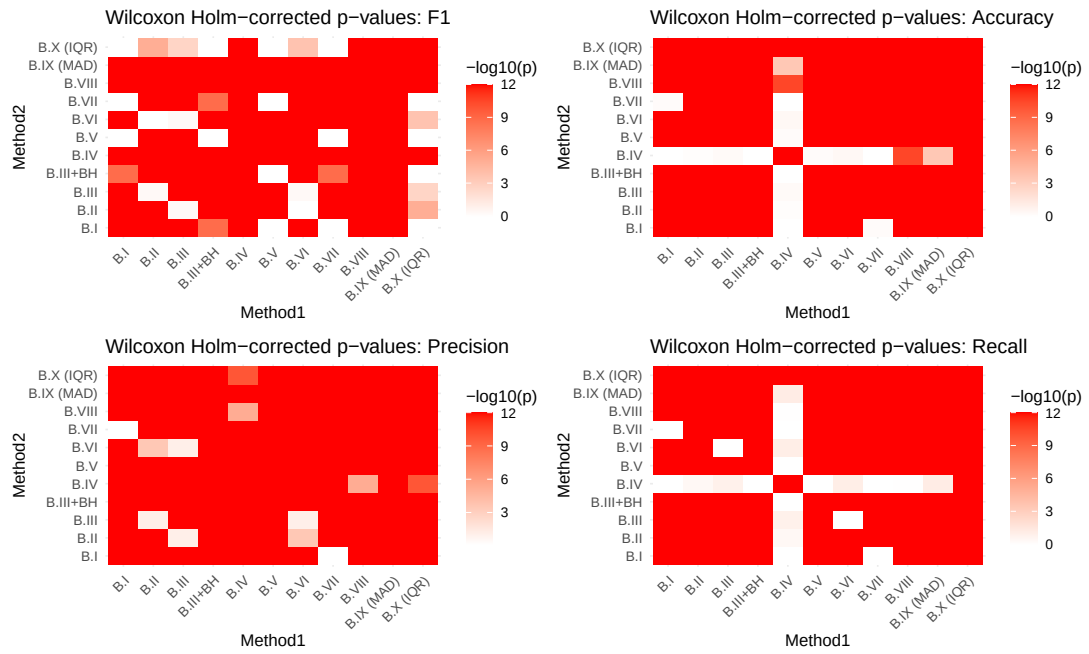


Figure 10: Adjusted p-values for Wilcoxon-Holm for B2, $n = 1000$, $p = 0.05$.

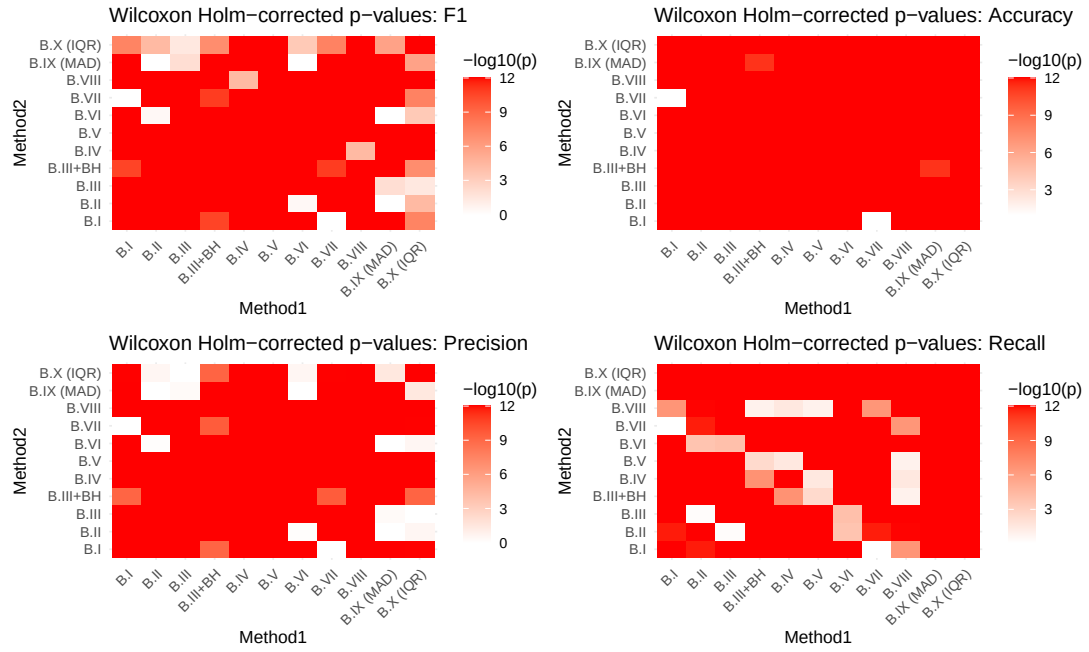


Figure 11: Adjusted p-values for Wilcoxon-Holm for B3, $n = 100$, $p = 0.05$.

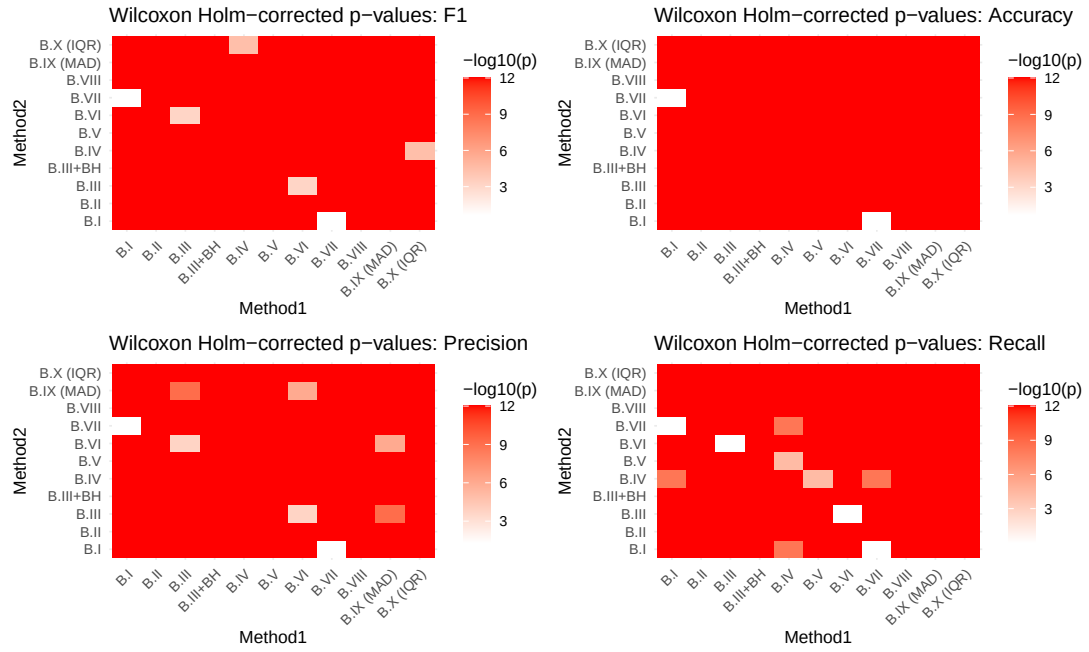


Figure 12: Adjusted p-values for Wilcoxon-Holm for B3, $n = 1000$, $p = 0.05$.

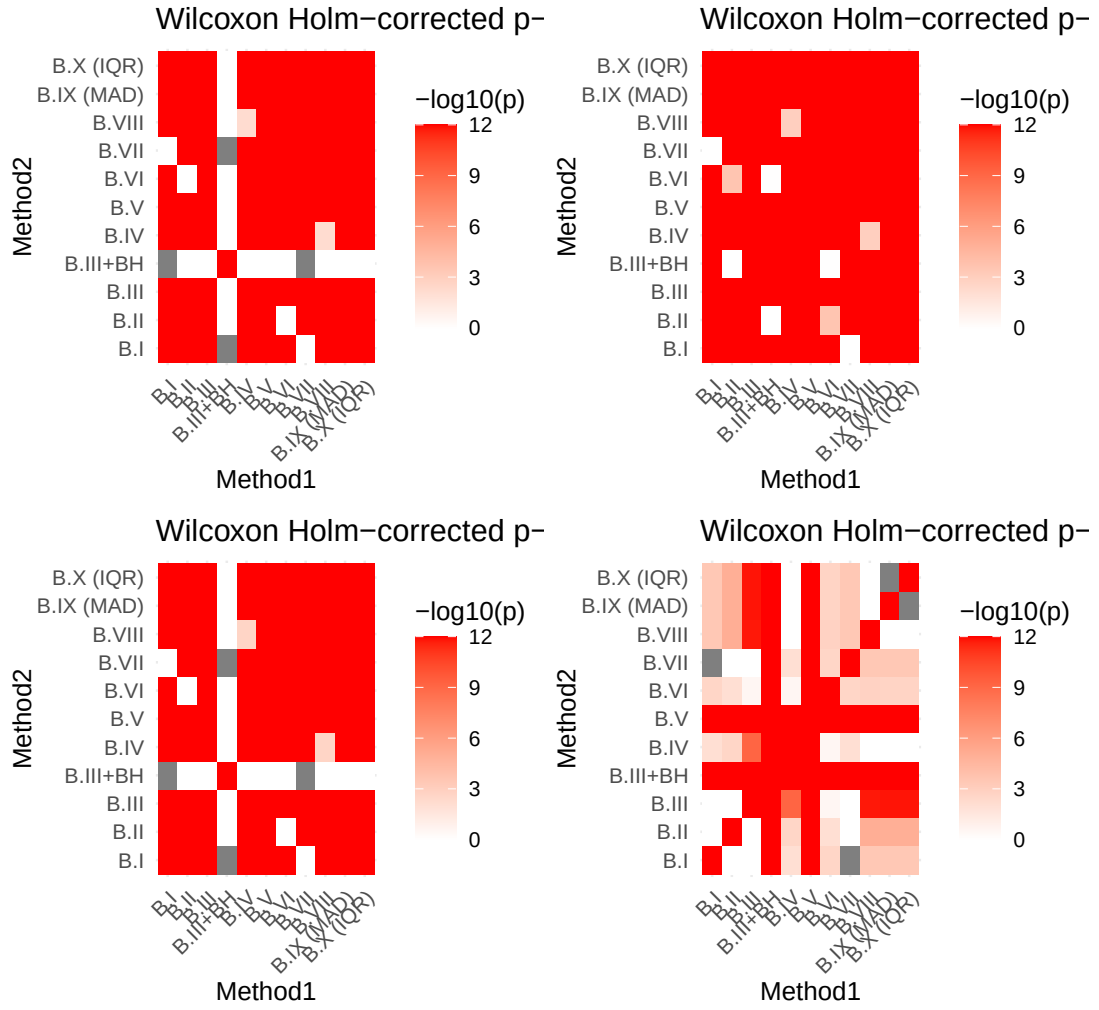


Figure 13: Adjusted p-values for Wilcoxon-Holm for A1, $n = 100$, $p = 0.1$.

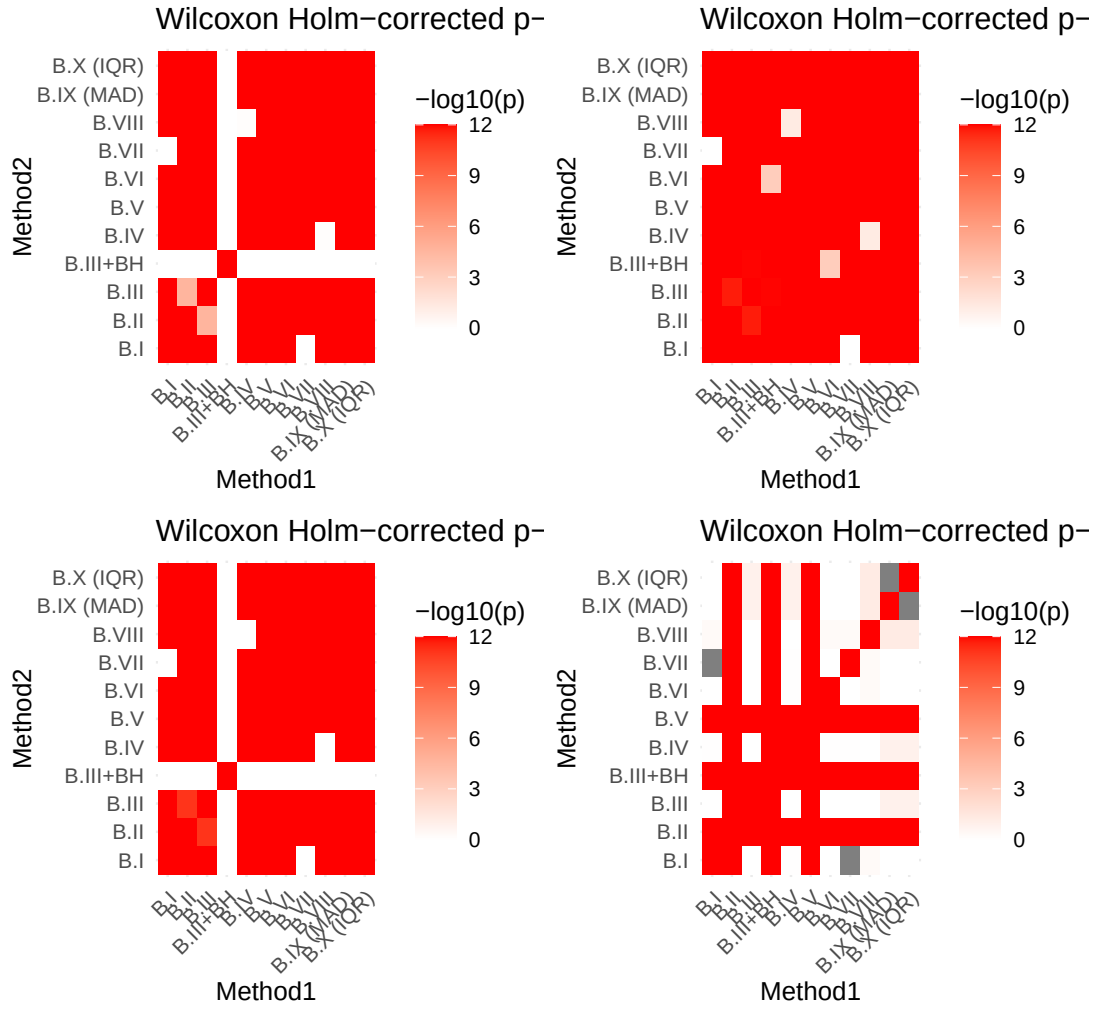


Figure 14: Adjusted p-values for Wilcoxon-Holm for A1, $n = 1000$, $p = 0.1$.

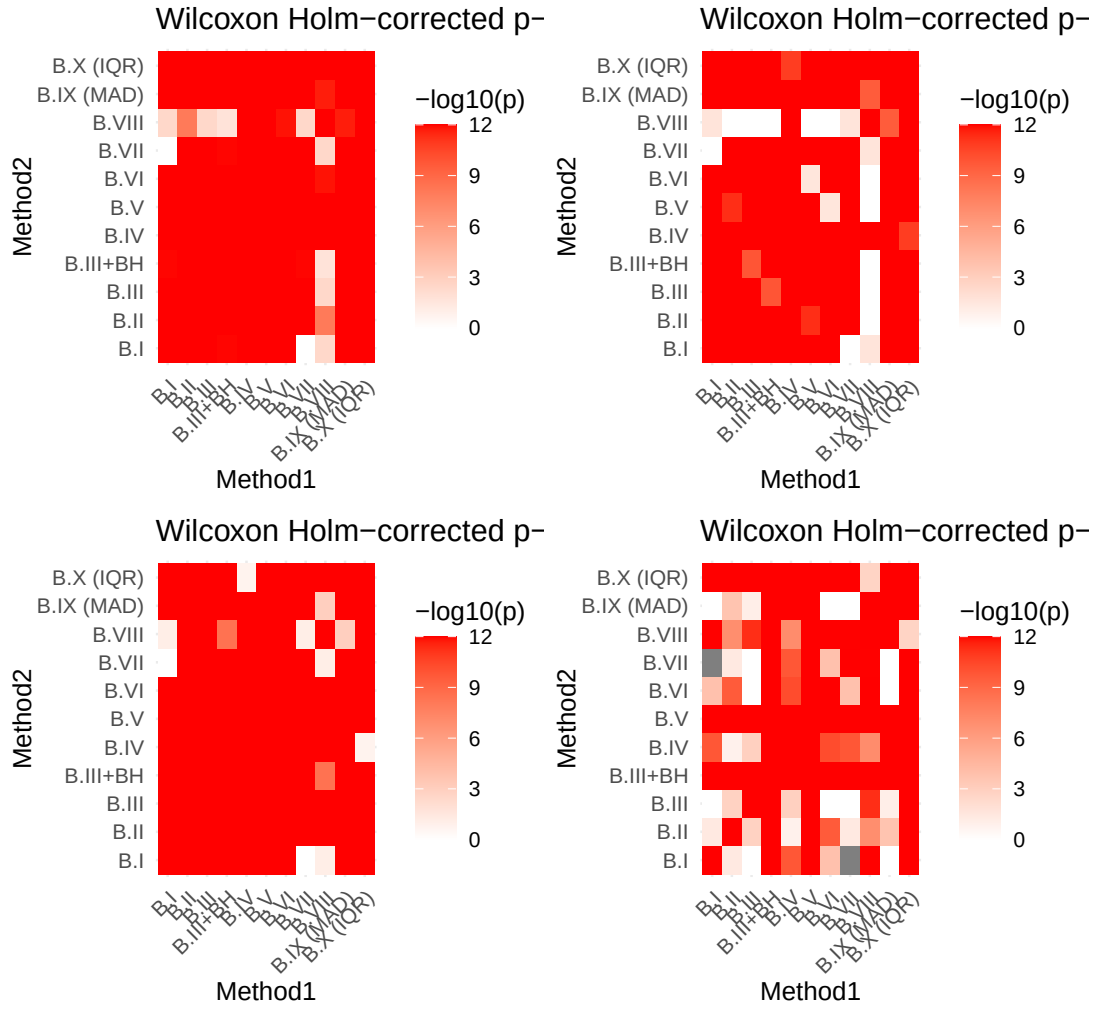


Figure 15: Adjusted p-values for Wilcoxon-Holm for A2, $n = 100$, $p = 0.1$.

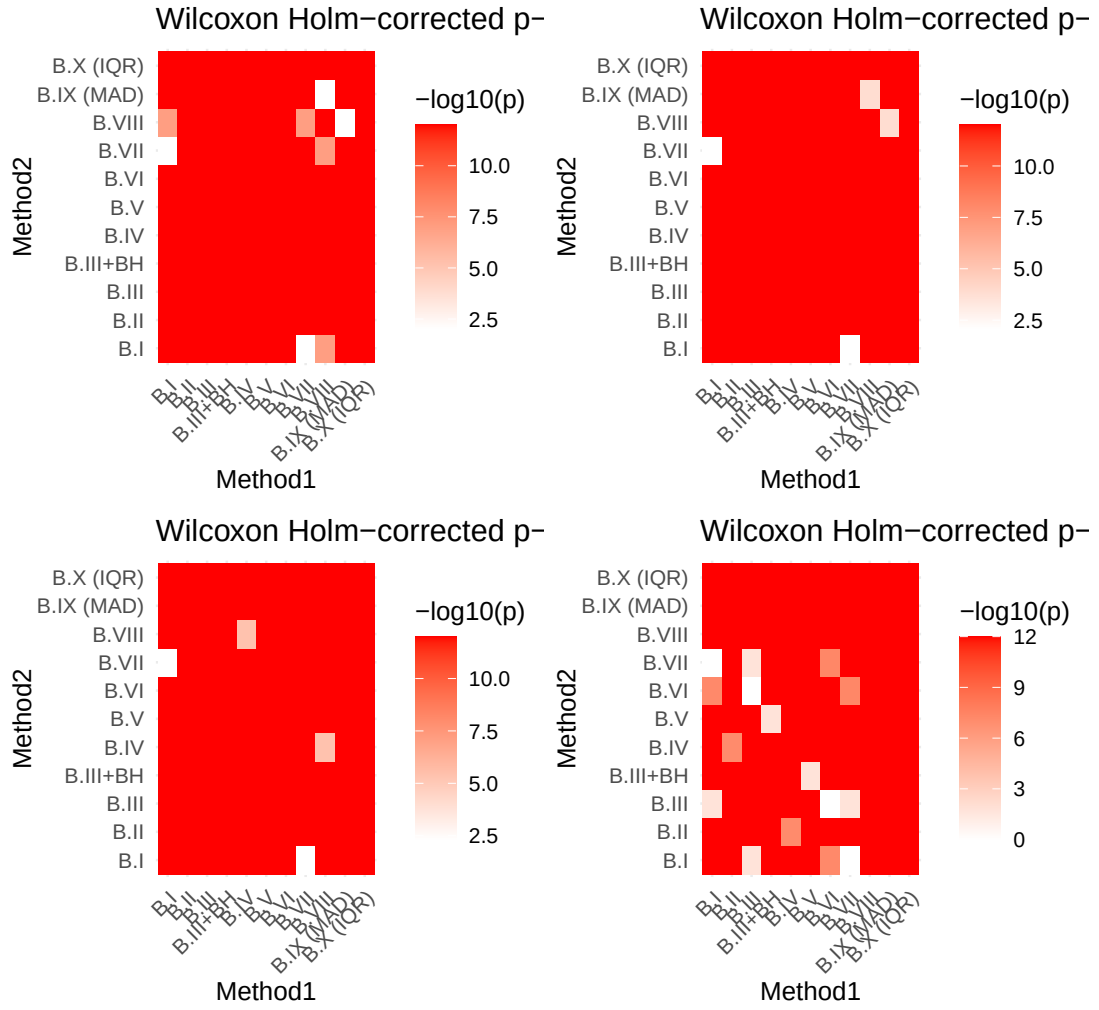


Figure 16: Adjusted p-values for Wilcoxon-Holm for A2, $n = 1000$, $p = 0.1$.

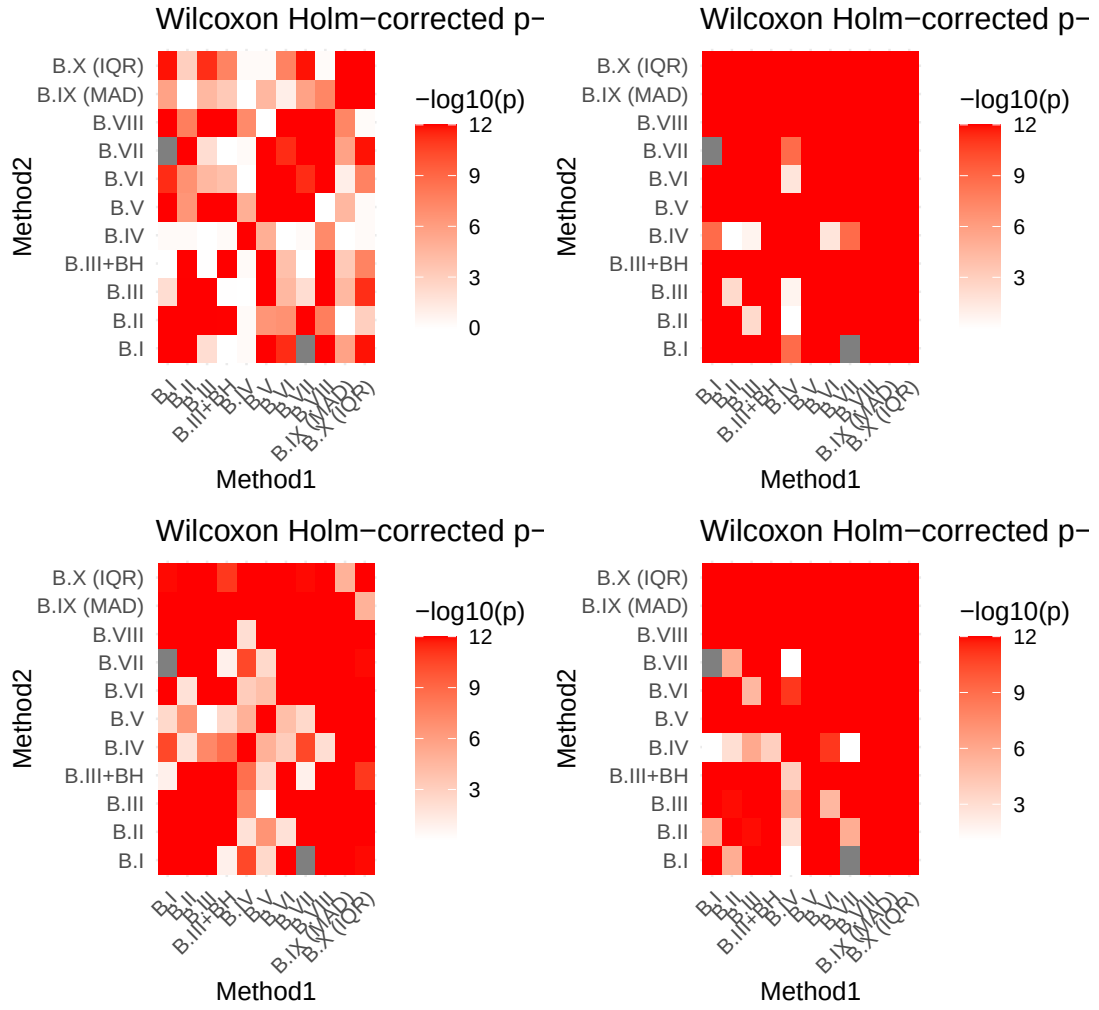


Figure 17: Adjusted p-values for Wilcoxon-Holm for A3, $n = 100$, $p = 0.1$.

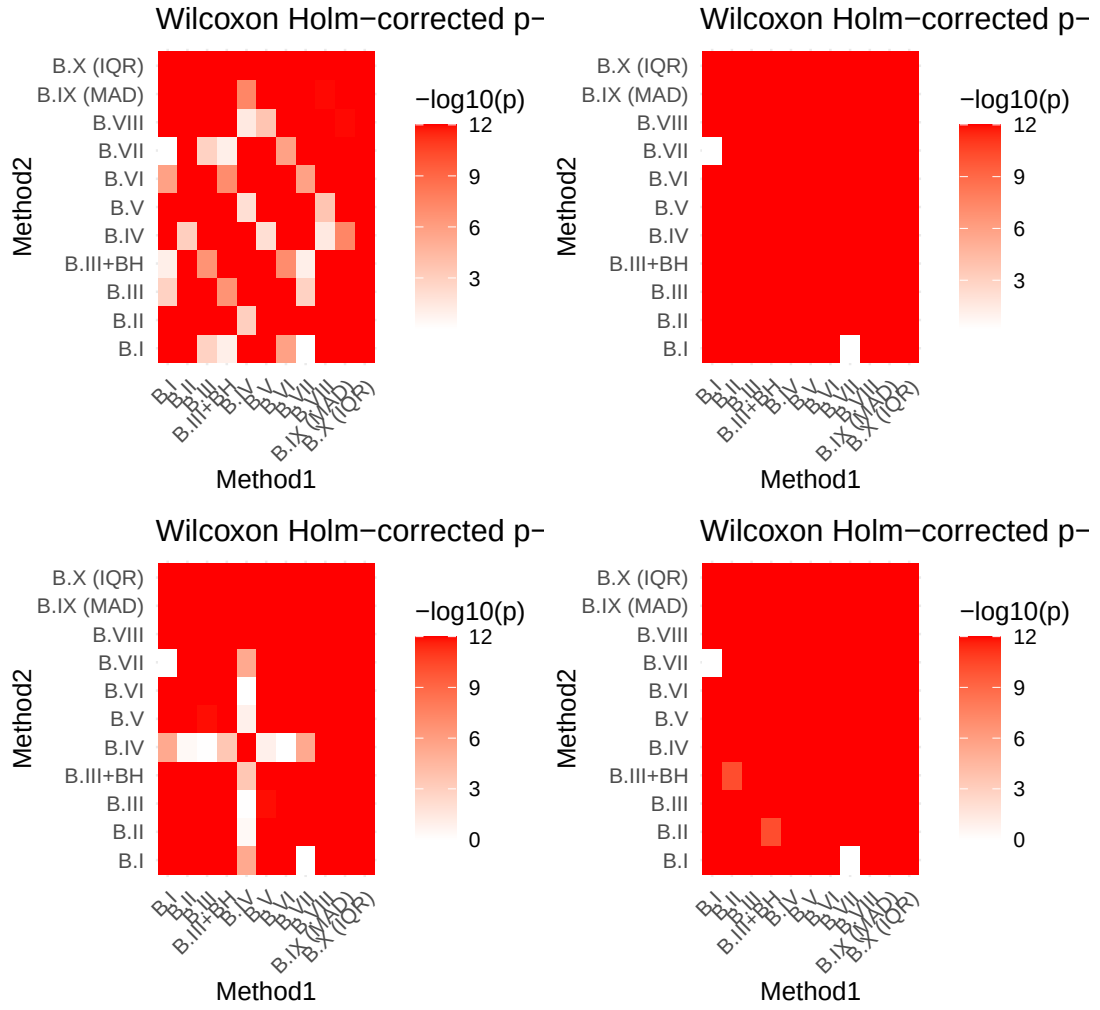


Figure 18: Adjusted p-values for Wilcoxon-Holm for A3, $n = 1000$, $p = 0.1$.

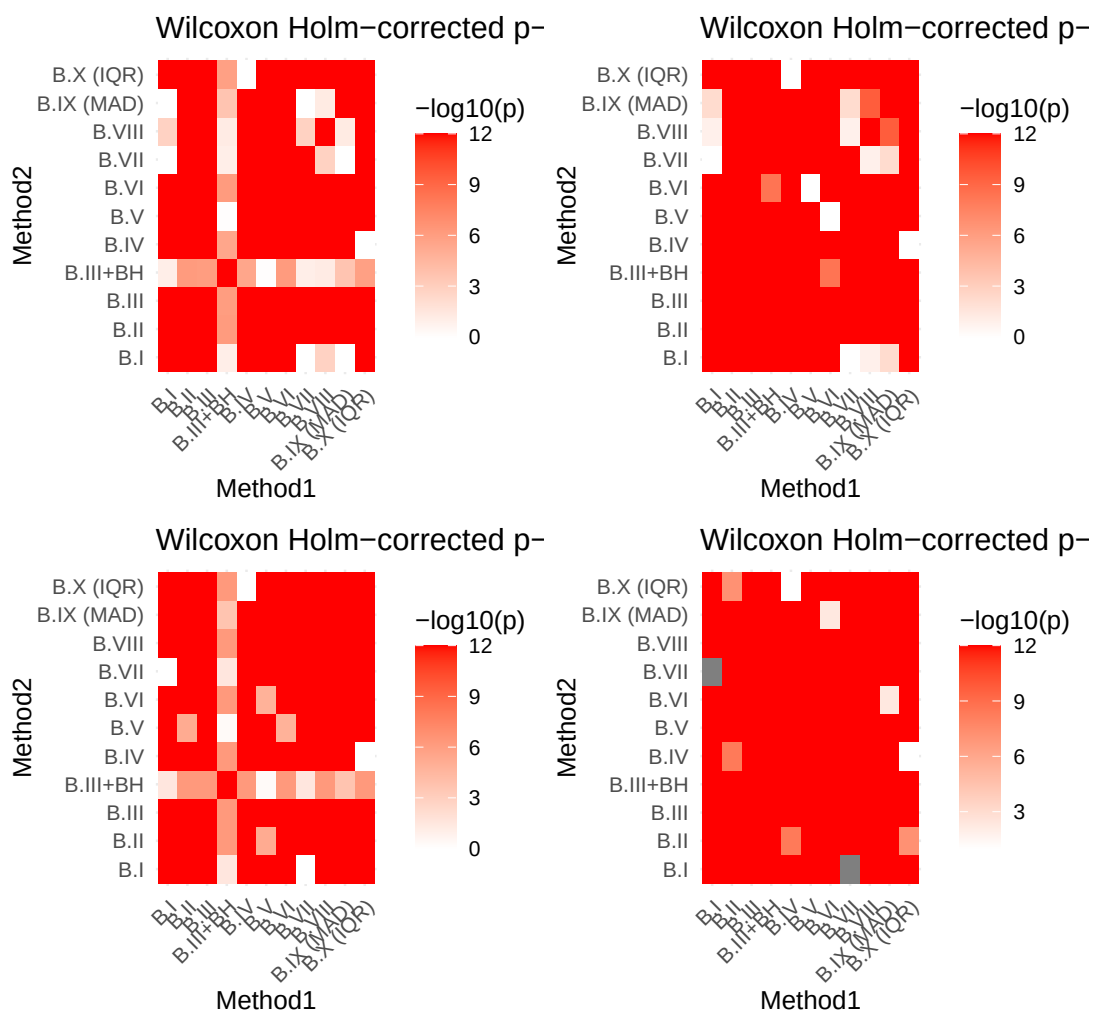


Figure 19: Adjusted p-values for Wilcoxon-Holm for B1, $n = 100$, $p = 0.1$.

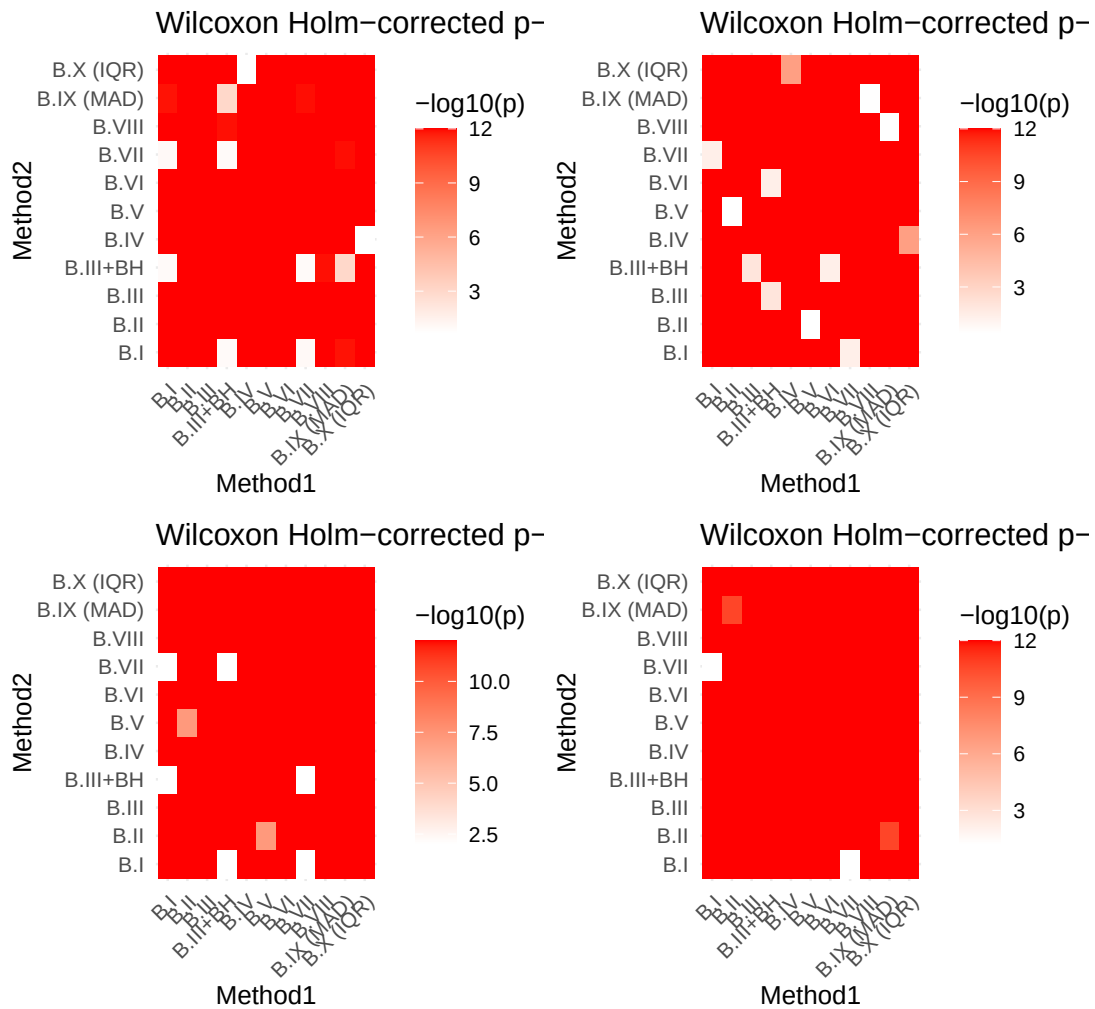


Figure 20: Adjusted p-values for Wilcoxon-Holm for B1, $n = 1000$, $p = 0.1$.

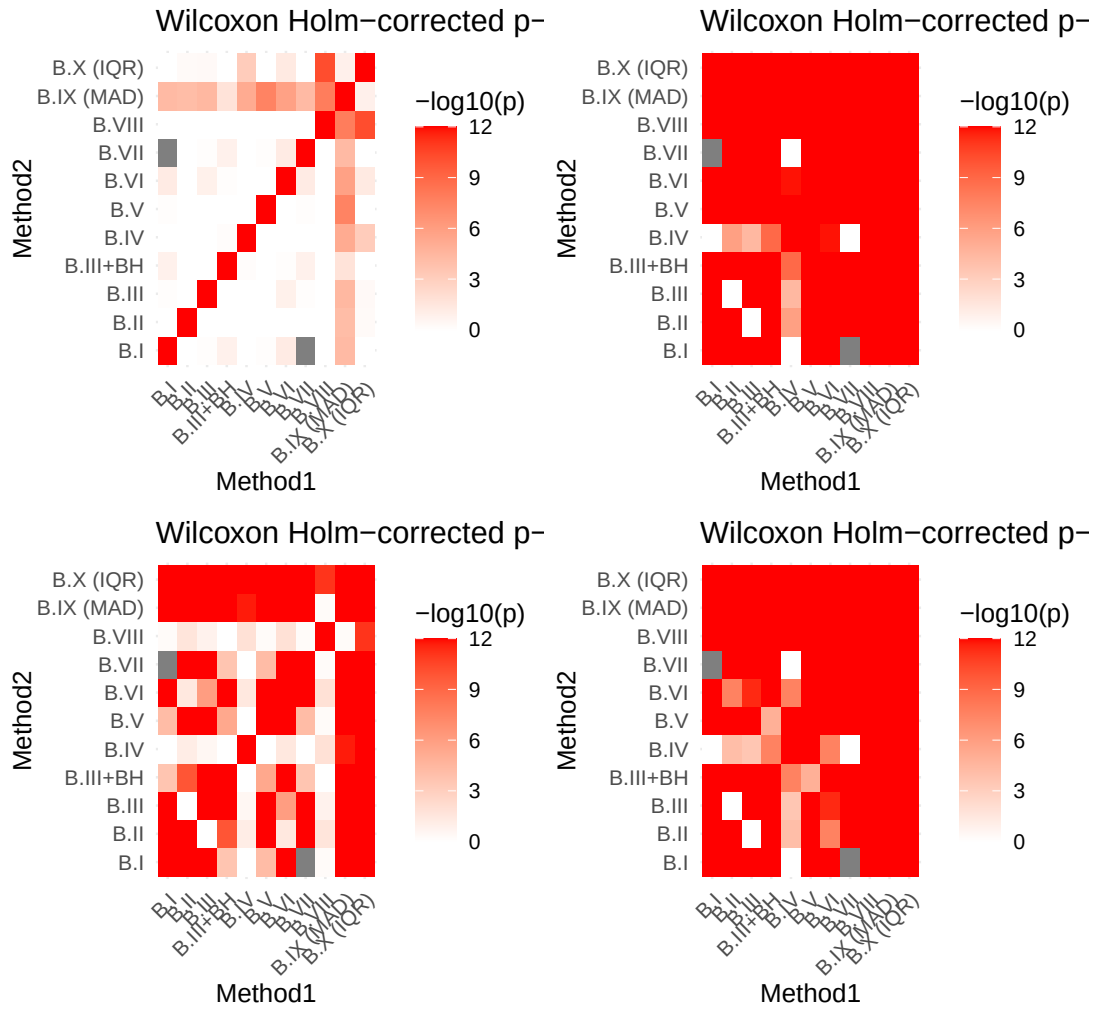


Figure 21: Adjusted p-values for Wilcoxon-Holm for B2, $n = 100$, $p = 0.1$.

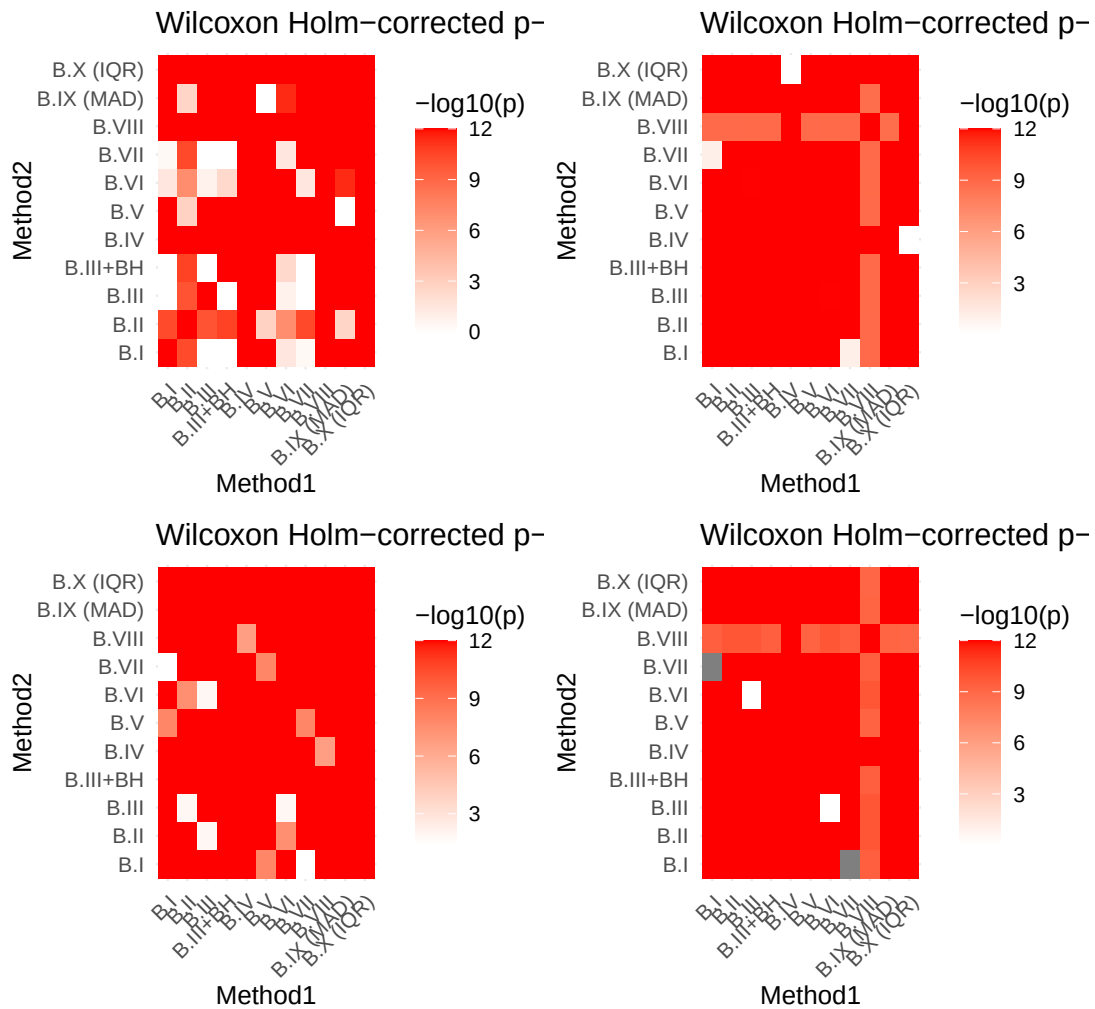


Figure 22: Adjusted p-values for Wilcoxon-Holm for B2, $n = 1000$, $p = 0.1$.

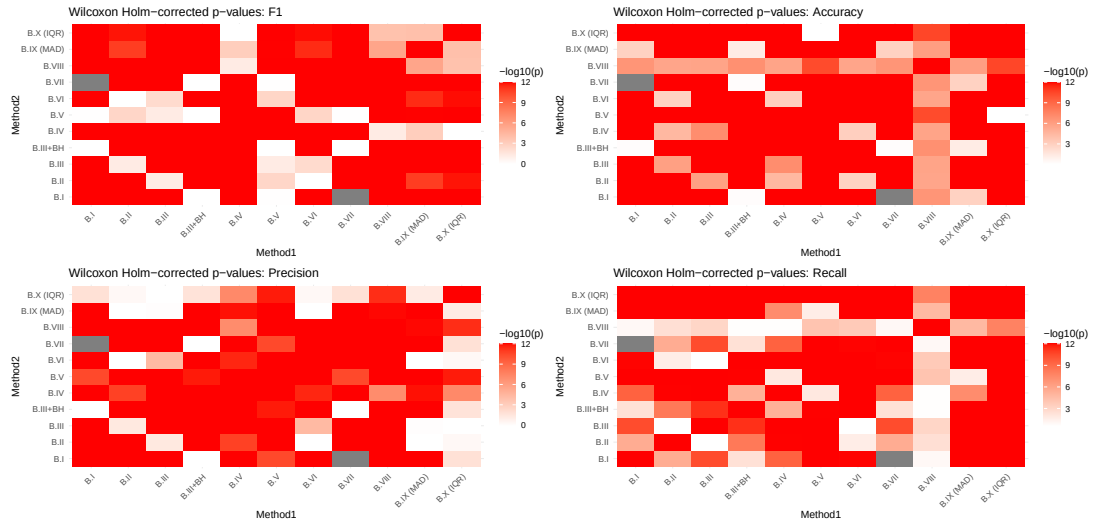


Figure 23: Adjusted p-values for Wilcoxon-Holm for B3, $n = 100$, $p = 0.1$.

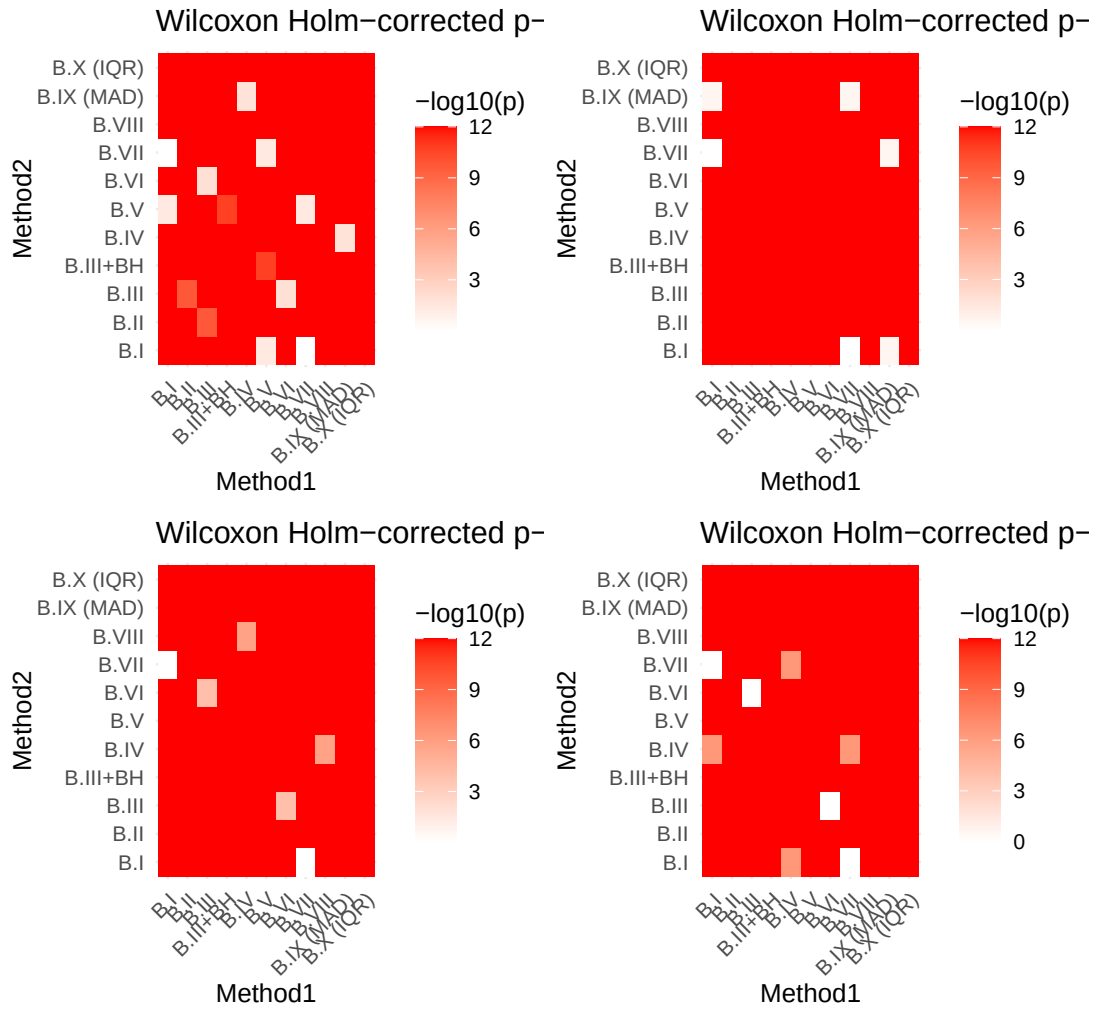


Figure 24: Adjusted p-values for Wilcoxon-Holm for B3, $n = 1000$, $p = 0.1$.

Results for the real-world data

Results for *Wine*

Feature	Method	Outliers	Percent (%)
flavanoids	M.I (KMeans+IF)	12	6.74
flavanoids	M.II (Naive Conformal)	7	3.93
flavanoids	M.III (Bates-like)	12	6.74
flavanoids	M.IIIb (Bates+BH)	12	6.74
flavanoids	M.IV (NormalmixEM)	36	20.22
flavanoids	M.V (OCSVM (all))	12	6.74
flavanoids	M.VI (OCSVM (inliers))	24	13.48
flavanoids	M.VII (LogReg)	12	6.74
flavanoids	M.VIII (NpEM)	0	0.00
flavanoids	M.IX (MAD)	1	0.56
flavanoids	M.X (IQR)	0	0.00
od280.od315_of_diluted_wines	M.I (KMeans+IF)	43	24.16
od280.od315_of_diluted_wines	M.II (Naive Conformal)	27	15.17
od280.od315_of_diluted_wines	M.III (Bates-like)	43	24.16
od280.od315_of_diluted_wines	M.IIIb (Bates+BH)	43	24.16
od280.od315_of_diluted_wines	M.IV (NormalmixEM)	54	30.34
od280.od315_of_diluted_wines	M.V (OCSVM (all))	14	7.87
od280.od315_of_diluted_wines	M.VI (OCSVM (inliers))	54	30.34
od280.od315_of_diluted_wines	M.VII (LogReg)	43	24.16
od280.od315_of_diluted_wines	M.VIII (NpEM)	0	0.00
od280.od315_of_diluted_wines	M.IX (MAD)	0	0.00
od280.od315_of_diluted_wines	M.X (IQR)	0	0.00
hue	M.I (KMeans+IF)	49	27.53
hue	M.II (Naive Conformal)	37	20.79
hue	M.III (Bates-like)	49	27.53
hue	M.IIIb (Bates+BH)	49	27.53
hue	M.IV (NormalmixEM)	18	10.11
hue	M.V (OCSVM (all))	24	13.48
hue	M.VI (OCSVM (inliers))	57	32.02
hue	M.VII (LogReg)	49	27.53
hue	M.VIII (NpEM)	0	0.00
hue	M.IX (MAD)	1	0.56
hue	M.X (IQR)	1	0.56
color_intensity	M.I (KMeans+IF)	42	23.60
color_intensity	M.II (Naive Conformal)	41	23.03
color_intensity	M.III (Bates-like)	42	23.60
color_intensity	M.IIIb (Bates+BH)	42	23.60
color_intensity	M.IV (NormalmixEM)	57	32.02
color_intensity	M.V (OCSVM (all))	11	6.18
color_intensity	M.VI (OCSVM (inliers))	58	32.58
color_intensity	M.VII (LogReg)	42	23.60
color_intensity	M.VIII (NpEM)	0	0.00
color_intensity	M.IX (MAD)	12	6.74
color_intensity	M.X (IQR)	4	2.25

total_phenols	M.I (KMeans+IF)	26	14.61
total_phenols	M.II (Naive Conformal)	26	14.61
total_phenols	M.III (Bates-like)	26	14.61
total_phenols	M.IIIb (Bates+BH)	26	14.61
total_phenols	M.IV (NormalmixEM)	69	38.76
total_phenols	M.V (OCSVM (all))	26	14.61
total_phenols	M.VI (OCSVM (inliers))	46	25.84
total_phenols	M.VII (LogReg)	26	14.61
total_phenols	M.VIII (NpEM)	0	0.00
total_phenols	M.IX (MAD)	1	0.56
total_phenols	M.X (IQR)	0	0.00

Results for *Heart Disease*

Feature	Method	Outliers	Percent (%)
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thalach	M.I (KMeans+IF)	412	40.20
thalach	M.II (Naive Conformal)	187	18.24
thalach	M.III (Bates-like)	94	9.17
thalach	M.IIIb (Bates+BH)	0	0.00
thalach	M.IV (NormalmixEM)	412	40.20
thalach	M.V (OCSVM (all))	89	8.68
thalach	M.VI (OCSVM (inliers))	509	49.66
thalach	M.VII (LogReg)	412	40.20
thalach	M.VIII (NpEM)	0	0.00
thalach	M.IX (MAD)	40	3.90
thalach	M.X (IQR)		

Results for Thyroid Disease

Feature	Method	Outliers	Percent (%)
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TSH	M.I (KMeans+IF)	22	0.58
TSH	M.II (Naive Conformal)	22	0.58
TSH	M.III (Bates-like)	22	0.58
TSH	M.IIIb (Bates+BH)	22	0.58
TSH	M.IV (NormalmixEM)	254	6.74
TSH	M.V (OCSVM (all))	153	4.06
TSH	M.VI (OCSVM (inliers))	156	4.14
TSH	M.VII (LogReg)	22	0.58
TSH	M.VIII (NpEM)	22	0.58
TSH	M.IX (MAD)	403	10.69
TSH	M.X (IQR)	258	6.84

FTI	M.I (KMeans+IF)	758	20.10
FTI	M.II (Naive Conformal)	644	17.08
FTI	M.III (Bates-like)	311	8.25
FTI	M.IIIb (Bates+BH)	9	0.24
FTI	M.IV (NormalmixEM)	561	14.88
FTI	M.V (OCSVM (all))	223	5.91
FTI	M.VI (OCSVM (inliers))	1323	35.08
FTI	M.VII (LogReg)	758	20.10
FTI	M.VIII (NpEM)	11	0.29

FTI M.IX (MAD) 440 11.67
 FTI M.X (IQR) 274 7.27

TT4 M.I (KMeans+IF) 950 25.19
 TT4 M.II (Naive Conformal) 755 20.02
 TT4 M.III (Bates-like) 318 8.43
 TT4 M.IIIb (Bates+BH) 8 0.21
 TT4 M.IV (NormalmixEM) 552 14.64
 TT4 M.V (OCSVM (all)) 194 5.14
 TT4 M.VI (OCSVM (inliers)) 1101 29.20
 TT4 M.VII (LogReg) 950 25.19
 TT4 M.VIII (NpEM) 5 0.13
 TT4 M.IX (MAD) 402 10.66
 TT4 M.X (IQR) 215 5.70

Results for *Credit Card Fraud Detection*

Feature Method Outliers Percent (%)

V17 M.I (KMeans+IF) 463 23.15
 V17 M.II (Naive Conformal) 397 19.85
 V17 M.III (Bates-like) 463 23.15
 V17 M.IIIb (Bates+BH) 463 23.15
 V17 M.IV (NormalmixEM) 58 2.90
 V17 M.V (OCSVM (all)) 99 4.95
 V17 M.VI (OCSVM (inliers)) 538 26.90
 V17 M.VII (LogReg) 463 23.15
 V17 M.VIII (NpEM) 6 0.30
 V17 M.IX (MAD) 141 7.05
 V17 M.X (IQR) 53 2.65

V14 M.I (KMeans+IF) 203 10.15
 V14 M.II (Naive Conformal) 199 9.95
 V14 M.III (Bates-like) 203 10.15
 V14 M.IIIb (Bates+BH) 203 10.15
 V14 M.IV (NormalmixEM) 41 2.05
 V14 M.V (OCSVM (all)) 101 5.05
 V14 M.VI (OCSVM (inliers)) 295 14.75
 V14 M.VII (LogReg) 203 10.15
 V14 M.VIII (NpEM) 1 0.05
 V14 M.IX (MAD) 186 9.30
 V14 M.X (IQR) 69 3.45

V10 M.I (KMeans+IF) 150 7.50
 V10 M.II (Naive Conformal) 105 5.25
 V10 M.III (Bates-like) 150 7.50
 V10 M.IIIb (Bates+BH) 150 7.50
 V10 M.IV (NormalmixEM) 47 2.35
 V10 M.V (OCSVM (all)) 103 5.15
 V10 M.VI (OCSVM (inliers)) 240 12.00
 V10 M.VII (LogReg) 150 7.50
 V10 M.VIII (NpEM) 14 0.70
 V10 M.IX (MAD) 211 10.55

V10 M.X (IQR) 65 3.25

V12 M.I (KMeans+IF) 278 13.90
V12 M.II (Naive Conformal) 230 11.50
V12 M.III (Bates-like) 278 13.90
V12 M.IIIb (Bates+BH) 278 13.90
V12 M.IV (NormalmixEM) 158 7.90
V12 M.V (OCSVM (all)) 101 5.05
V12 M.VI (OCSVM (inliers)) 362 18.10
V12 M.VII (LogReg) 278 13.90
V12 M.VIII (NpEM) 0 0.00
V12 M.IX (MAD) 174 8.70
V12 M.X (IQR) 83 4.15

V11 M.I (KMeans+IF) 140 7.00
V11 M.II (Naive Conformal) 89 4.45
V11 M.III (Bates-like) 140 7.00
V11 M.IIIb (Bates+BH) 140 7.00
V11 M.IV (NormalmixEM) 912 45.60
V11 M.V (OCSVM (all)) 98 4.90
V11 M.VI (OCSVM (inliers)) 232 11.60
V11 M.VII (LogReg) 140 7.00
V11 M.VIII (NpEM) 1 0.05
V11 M.IX (MAD) 37 1.85
V11 M.X (IQR) 4 0.20
